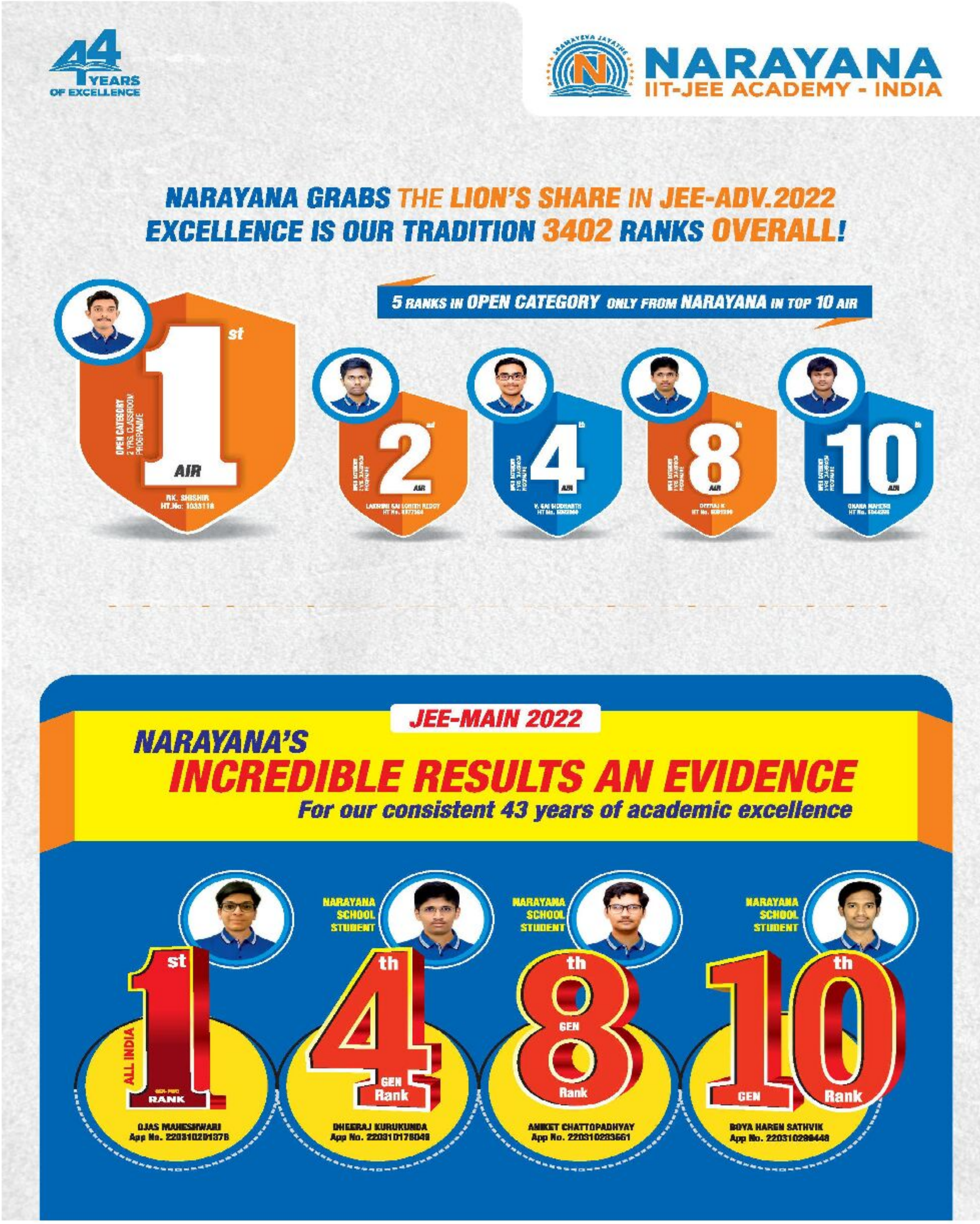


MATHS

VECTORS & 3D

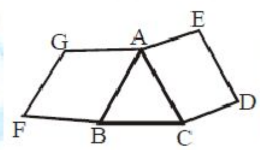
JEE ADVANCED IMPORTANT QUESTIONS

SINGLE CORRECT

1. If $\left[\bar{a}\,\bar{b}\,\bar{c}\right] = |\bar{d}| = 6$, then $\left|\left[\bar{d}\,\bar{b}\,\bar{c}\right]\bar{a} + \left[\bar{d}\,\bar{c}\,\bar{a}\right]\bar{b} + \left[\bar{d}\,\bar{a}\,\bar{b}\right]\bar{c}\right| =$

(A) 36 (B) 12 (C) 24 (D) 6

2. In the given figure ABFG is parallelogram and ACDE is a square. If $\overrightarrow{BC} = \vec{a}, \overrightarrow{FG} = \vec{b}, \overrightarrow{CE} = \vec{c}, \overrightarrow{BG} = \vec{a} - \vec{b}$ and if $\overrightarrow{GE} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c}$ then $(\alpha + \beta + \gamma)$ is ___



(A) 2 (B) –2 (C) 1 (D) 0

3. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be 3 mutually perpendicular unit vectors. If $\vec{d}$ is a unit vector which is equally inclined to $\vec{a}, \vec{b}$ and $\vec{c}$, then the value of $\left|\vec{a} + \vec{b} + \vec{c} + \vec{d}\right|^2$ is ____

(A) $4 + 2\sqrt{3}$ (B) $4 + 3\sqrt{2}$ (C) $2 + \sqrt{5}$ (D) $\sqrt{5} - 2$

4. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar unit vector, each inclined with other at angle $\frac{\pi}{6}$, then volume of tetrahedron whose edge are $\vec{a}, \vec{b}$ and $\vec{c}$ is

(A) $\frac{\sqrt{3\sqrt{3} - 5}}{12}$ (B) $\frac{2\sqrt{3} + 5}{12}$ (C) $\frac{3 + 5\sqrt{2}}{12}$ (D) None

5. Let $|\vec{a}| = 1, |\vec{b}| = 2, |\vec{c}| = 3$ where $\vec{a} \wedge \vec{b} = 60°$ & $\vec{c}$ bisects the angle between $\vec{a}$ & $\vec{b}$ and $\left|\vec{a} + \vec{b} + \vec{c}\right| = \left(\ell + m\sqrt{3}\right)$ where $\ell, m \in N$ then $(\ell + m)$ is equal to

(A) 5 (B) 15 (C) 25 (D) 35

6. Given vectors $\vec{a} = \sin\frac{\pi}{6}\hat{i} + \sin\frac{\pi}{3}\hat{j}; \vec{b} = \sin\frac{\pi}{3}\hat{j} + \sin\frac{\pi}{6}\hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$. The volume of parallelopiped with three conterminous edges as

$\vec{u} = (\vec{a}.\vec{a})\vec{a} + (\vec{a}.\vec{b})b + (\vec{a}.\vec{c})\vec{c}$

$\vec{v} = (\vec{a}.\vec{b})\vec{a} + (\vec{b}.\vec{b})\vec{b} + (\vec{b}.\vec{c})\vec{c}$

$\vec{w} = (\vec{a}.\vec{c})\vec{a} + (\vec{b}.\vec{c})\vec{b} + (\vec{c}.\vec{c})\vec{c}$ equals

(A) $\left(2\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\right)^3$ (B) $\left(2\cos\frac{\pi}{3} - \sin\frac{\pi}{3}\right)^3$

(C) $\left(2\cos\frac{\pi}{6} - \sin\frac{\pi}{6}\right)^3$ (D) $\left(2\sin\frac{\pi}{4} - \cos\frac{\pi}{3}\right)^3$

7. A particle moves over a path, whose position vector at any time t, is given by $\vec{r}(t) = |t|\hat{i} + |\sin \pi t|\hat{j} + \sqrt[3]{t-2}\hat{k}$. A path $\vec{r}(t)$ is called differentiable at time t if its all components are differentiable with respect to 't' at that point of time. So $\vec{r}(t)$ is differentiable at t =

(A) 3 (B) 1 (C) 2 (D) 3.5

8. The equation of line which intersect each of two lines $2x + y - 1 = 0 = x - 2y + 3z$ and $3x - y + z + 2 = 0,\ 4x + 5y - 2z - 3 = 0$ and is parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is

(A) $4x + 7y - 6z - 3 = 0, 2x - 7y + 4z + 7 = 0$ (B) $4x + 7y - 6z - 4 = 0, 2x - 7y + 4z + 2 = 0$

(C) $4x + 7y - 6z - 2 = 0, 2x - 7y + 4z + 5 = 0$ (D) None of these

9. Equation of lines

$L_1 : 2x - 2y + 3z - 2 = 0, x - y + z + 1 = 0$

$L_2 : x + 2y - z - 3 = 0, 3x - y + 2z - 1 = 0$

Distance of point P(0, 0, 0) from the plane containing L_1 and L_2 and measured along the line $x = y = z$ is –

(A) $\frac{1}{3\sqrt{2}}$ (B) $\frac{3\sqrt{3}}{8}$ (C) $\frac{\sqrt{3}}{8}$ (D) $\frac{\sqrt{3}}{4}$

10. A line with direction cosines proportional to 2, 1, 2 meet the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ at P and Q respectively. The distance between P and Q is:

(A) 2 (B) 3 (C) 6 (D) 8

11. Let P be the image of the point (1, 2, 3) with respect to the plane $x - y + z + 1 = 0$. Then the equation of the plane passing through P and containing the line $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-7}{1}$ is

(A) $x + y - 3z + 17 = 0$ (B) $15x - 2y - 11z + 34 = 0$

(C) $x - 4y + 7z = 48$ (D) $x - y + z = 4$

12. If line x = y = z intersect the line $x\sin A + y\sin B + z\sin C - 2d^2 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - d^2$ then $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ equals, where $(A + B + C) = \pi$

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{32}$ (D) $\frac{1}{12}$

MULTI CORRECT

13. The vertices of a triangle ABC are $A = (2,0,2), B(-1,1,1)$ and $C = (1,-2,4)$.The points D and E divide the sides AB and CA in the ration 1:2 respectively. Another point F is taken in space such that the perpendicular drawn from F to the plane containing ΔABC ,meets the plane at the point of intersection of the line segments CD and BE. If the distance of F from the plane of triangle ABC is $\sqrt{2}units$,then

(A) the volume of the tetrahedron ABCF is $\frac{7}{3}$ cubic units

(B) the volume of the tetrahedron ABCF is $\frac{7}{6}$ cubic units

(C) one of the equation of the line AF is $\vec{r}.(2\hat{i} + 2\hat{k}) + \lambda(2\hat{k} - \hat{i})(\lambda \in R)$

(D) one of the equation of the line AF is $\vec{r}.(2\hat{i} + 2\hat{k}) + \mu(\hat{i} + 7\hat{k})$

14. The direction cosines of two lines are connected by the relation $\ell + m + n = 0$ and $mn = 2\ell(m+n)$ then,

(A) $\frac{\ell_1}{\ell_2} + \frac{m_1}{m_2} + \frac{n_1}{n_2}$ is equal to $-\frac{3}{2}$ (B) $\ell_1\ell_2 + m_1m_2 + n_1n_2$ is equal to $-\frac{1}{2}$

(C) $\ell_1m_1n_1 + \ell_2m_2n_2$ is equal to $-\frac{\sqrt{2}}{3\sqrt{3}}$ (D) $(\ell_1 + \ell_2)(m_1 + m_2)(n_1 + n_2)$ is equal to $\frac{1}{3\sqrt{6}}$

15. Which of the following is/are correct?

(A) The point of intersection of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ where $A(4,7,8), B(-1,-2,1), C(2,3,4)$ and $D(1,2,5)$ is $\left(\frac{3}{2},\frac{5}{2},\frac{9}{2}\right)$

(B) If (α, β, γ) be point on plane $2x + 6y + 12z = 7$ then the least value of $\alpha^2 + \beta^2 + 25\gamma^2$ is 7

(C) The circumcentre of the triangle formed by the points $(3,2,-5), (-3,8,-5)$ and $(-3,2,1)$ is the point $(-1,4,-3)$

(D) The length of the intercept on the line $\frac{x+1}{-1}=\frac{y-12}{5}=\frac{z-7}{2}$ by the surface $11x^2-5y^2+z^2=0$ is $\sqrt{30}$ *units*

16. Let $\vec{a},\vec{b},\vec{c}$ be vectors such that $\vec{b}.\vec{c}=2\vec{a}.\vec{a}=3\vec{b}.\vec{b}=\vec{a}.\vec{b}\times\vec{c}=1$ and $\vec{a}.\vec{b}=\vec{a}.\vec{c}=0$. Now if $\vec{a}+\vec{b}$ is expressed as $x\vec{a}\times y\vec{b}+z\vec{c}$, then

(A) $x<0$ (B) $y<0$ (C) $z<0$ (D) $x+y+z<0$

17. Two perpendicular unit vectors $\vec{a}$ and $\vec{b}$ are such that $\left[\vec{r}\ \vec{a}\ \vec{b}\right]=\frac{5}{4}$, $\vec{r}.\left(3\vec{a}+2\vec{b}\right)=0$ and $\int_{-2\vec{r}.\vec{a}}^{-\frac{4}{3}\vec{r}.\vec{b}}\frac{x+1}{x^2+1}dx=\frac{\pi}{2}$. Which of the following is (are) correct?

(A) $\left|\vec{r}\right|^2=\frac{19}{8}$ (B) $\left|\vec{r}\right|=\frac{7}{4}$

(C) $\vec{r}=\frac{\vec{a}}{2}-\frac{3\vec{b}}{4}+\frac{5}{4}\left(\vec{a}\times\vec{b}\right)$ (D) $\vec{r}=\frac{-\vec{a}}{2}+\frac{3\vec{b}}{2}+\frac{5}{4}\left(\vec{a}\times\vec{b}\right)$

18. Given $\vec{A}(x)=x\hat{i}+x\hat{j}+5\hat{k}$, $\vec{B}(x)=\hat{i}+x\hat{j}$, $\vec{C}(x)=\hat{i}+\frac{1}{5}\hat{j}+x\hat{k}$. Let α,β,γ denotes three values of x for which $\vec{A}(x),\vec{B}(x)$ and $\vec{C}(x)$ are coplanar, then

(A) $[\alpha]+[\beta]+[\gamma]=-1$ (B) $[\alpha]+[\beta]+[\gamma]=0$

(C) Global maximum value of volume of parallelepiped formed by three vectros whose coterminous edges are $\vec{A}(x),\vec{B}(x)\&\vec{C}(x)$ is 4 for $x\in[-2,3]$

(D) $(1+\alpha)(1+\beta)(1+\gamma)=-4$. (where [.] denotes greatest integer function)

19. Let position vectors of points A, B, C are $\vec{a},\vec{b},\vec{c}$ respectively where $\vec{c}=x\vec{a}+y\vec{b}$

(A) If $x+y=1, x>0, y<0$ then C lies on line AB but not on line segment AB

(B) If $x>0, y>0$, $x+y<1$ then C lies inside the triangle ΔABO

(C) If $x<0, y>0$ and $x+y>1$ then on plane containing points O, A, B, the points C and origin lies on opposite side of line AB.

(D) If $x\le 0$, $y\le 0$ and $x+y=-1$ then locus of point C is reflection of the line segment AB about origin

20. Let ABC be a triangle and a, b and c are sides length opposite to vertices A, B and C respectively. If $\vec{\alpha}$ and $\vec{\beta}$ are two non collinear unit vectors satisfying $(a-b)\vec{\alpha}+(b-c)\vec{\beta}+(c-a)(\vec{\alpha}\times\vec{\beta})=\vec{0}$, then ΔABC is

(A) An acute angled triangle
(B) An obtuse angled triangle
(C) A right angled triangle
(D) R = 2r (where R and r are circum radius and in-radius of ΔABC)

21. The value of $\left[\vec{a}\times\vec{p}\ \ \vec{b}\times\vec{q}\ \ \vec{c}\times\vec{r}\right]+\left[\vec{a}\times\vec{q}\ \ \vec{b}\times\vec{r}\ \ \vec{c}\times\vec{p}\right]+\left[\vec{a}\times\vec{r}\ \ \vec{b}\times\vec{p}\ \ \vec{c}\times\vec{q}\right]$ is

(A) independent of $\vec{a},\vec{b}$ and $\vec{c}$ (B) independent of $\vec{p},\vec{q}$ and $\vec{r}$
(C) 0 (D) dependent of $\vec{a},\vec{b}$ and $\vec{c}$

22. A non-zero vector $\vec{p}$ is equally inclined to three vectors $\vec{a}=\hat{i}-\hat{j}+\hat{k}, \vec{b}=2\hat{i}+\hat{j}, \vec{c}=3\hat{i}-2\hat{k}$. Let $\vec{x},\vec{y},\vec{z}$ be three vectors in plane of $\vec{a},\vec{b};\vec{b},\vec{c}$ and $\vec{c}$, $\vec{a}$ respectively then

(A) $\vec{x}.\vec{p}=1$ (B) $\vec{y}.\vec{p}=0$ (C) $\vec{z}.\vec{p}=0$ (D) $\left(2\hat{i}-3\hat{j}+4\hat{k}\right).\vec{p}=0$

23. A variable plane which remains at a constant distance P from the origin (0) cuts the coordinate axes in A, B, C.

(A) Locus of centroid tetrahedron OABC is $x^2y^2+y^2z^2+z^2x^2=\frac{16}{p^2}x^2y^2z^2$

(B) Locus of centroid tetrahedron OABC is $x^2y^2+y^2z^2+z^2x^2=\frac{4}{p^2}x^2y^2z^2$

(C) Parametric equation of the centroid of the tetrahedron is of the form $\left(\frac{P}{4}\sec\alpha\sec\beta,\frac{P}{4}\sec\alpha\operatorname{cosec}\beta,\frac{P}{4}\operatorname{cosec}\alpha\right)\alpha,\beta\in(0,2\pi)-\{\pi/2,\pi,3\pi/2\}$

(D) None of these

24. Three mutually perpendicular lines are drawn from the point $(1,2,-1)$. If one of the lines is perpendicular to the x-axis and the direction ratios of the second line are $(1,2,-1)$, then the equations(s) of the third line is/are

(A) $\frac{x-1}{5}=\frac{y-2}{-2}=\frac{z+1}{1}$ (B) $\frac{x+4}{5}=\frac{y-4}{-2}=\frac{z+2}{1}$

(C) $\vec{r}=6\hat{i}+\lambda\left(5\hat{i}-2\hat{j}+\hat{k}\right)$ (D) $\frac{x+1}{5}=\frac{y+2}{-2}=\frac{z-1}{1}$

25. Consider Lines $L_1 : \frac{x-\alpha}{1} = \frac{y}{-2} = \frac{z+p}{2}, L_2 : x = \alpha, \frac{y}{-\alpha} = \frac{z+\alpha}{2-\beta}$,plane $P : 2x+2y+z+7=0$. Let the; line L_2 lies in plane P, then

(A) $\alpha = -7$

(B) $\alpha = 7$

(C) minimum distance between line L_1 and plane P is 11.

(D) minimum distance between line L_1 and plane P is $\frac{23}{3}$

26. Consider the lines $L_1 : \frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $L_2 : \frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ then the line along shortest distance can be, constituted by the line of intersection of planes

(A) $4x+y-5z=0$ (B) $x-3y+5z=0$

(C) $5x-7y+2z=-1$ (D) $7x+y-8z=31$

27. Let $L_1 : x+y=1, z+2y=2$ and $L_2 : x-y=1, z-2y=2$ be two straight lines, then which of the following is/are correct statement(s)?

(A) the lines L_1 and L_2 are coplanar

(B) the lines L_1 and L_2 are non-coplanar

(C) on the line $\vec{r} = 4\hat{j} + \lambda\left(\hat{i} + 2\hat{k}\right), \lambda \in R$, there exist infinite number of points such that an infinite number of lines passing through each of these points and intersecting both the lines L_1 and L_2 can be drawn

(D) there exists a unique line passing through the point $A(1, \mu, 3)$ which intersect both lines L_1 and L_2 $\forall \mu \in R$

28. Let E-ABCD be a pyramid on square base ABCD where A is the origin and B and D are lying on positive x-axis and y-axis respectively. If E is (0, 2, 3) and $\overline{DE}.(\hat{i} + \hat{j}) = \vec{0}$, then

(A) image of the point D in the plane ABE is $\left(0, \frac{-10}{13}, \frac{24}{13}\right)$

(B) image of the point D in the plane ABE is $\left(0, \frac{-6}{13}, \frac{30}{13}\right)$

(C) volume of the tetrahedron ABDE is 2 cubic units.

(D) perpendicular distance of the point D from the plane ABE is $\frac{9}{\sqrt{13}}$

29. Consider points A (0,0,0), B(1, 1, 1) and plane P : $x - y + z = 3$. Let Q(a,b,c) be a moving point on plane P then identify the correct statement(s)

(A) Uf QA + QB is minimum then $a - c - 10b$ is equal to 1

(B) If QA + QB is minimum then $a - c- 10b$ is equal to 2

(C) If $|QA - QB|$ is maximum then $a - 2b + c$ is equal to 0

(D) If $|QA - QB|$is minimum then Q is unique

30. If (α, β, γ) is a point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at distance $2\sqrt{7}$ from the point $(1,-1,0)$, then the coefficient of $x^{\alpha+\beta+\gamma}$ in the expansion of $\left(2x^2 - \frac{1}{x}\right)^9$ is divisible by

(A) 21 (B) 36 (C) 28 (D) 24

31. If the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ intersects line $3\beta^2 x + 3(1-2\alpha)y + z = 3 = -\frac{1}{2}\left(6\alpha^2 x + 3(1-2\beta)y + 2z\right)$ then point $(\alpha, \beta, 1)$ lies on the plane

(A) $2x - y + z = 4$ (B) $x + y - z = 2$

(C) $x - 2y = 0$ (D) $2x - y = 0$

32. Consider the planes $P_1 : 2x + y + z + 4 = 0, P_2 : y - z + 4 = 0$ and $P_3 : 3x + 2y + z + 8 = 0$. Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3, P_3 and P_1, and P_1 and P_2 respectively. Then,

(A) Atleast two of the lines L_1, L_2 and L_3 are non-parallel

(B) Atleast two of the lines L_1, L_2 and L_3 are parallel

(C) The three planes intersect in a line

(D) The three planes form a triangular prism

33. If the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ intersects the line

$3\beta^2 x + 3(1-2\alpha)y + z = 3 = -\frac{1}{2}\left\{6\alpha^2 x + 3(1-2\beta)y + 2z\right\}$, then point $(\alpha, \beta, 1)$ lie on the plane

(A) $2x - y + z = 4$ (B) $x + y - z = 2$ (C) $x - 2y = 0$ (D) $2x - y = 0$

INTEGER TYPE QUESTIONS

34. Let 3 vector $\vec{\ell}, \vec{m}$ & $\vec{n}$ are such that $|\vec{\ell}| = 3, |\vec{m}| = \frac{1}{2}$ & $\vec{\ell}.\vec{m} = \vec{m}.\vec{n} = \vec{n}.\vec{\ell} = 0$. If $\left[\vec{\ell} \times \left(\vec{m} \times \vec{\ell}\right) \vec{n} \times \vec{m} \ \vec{n}\right] = 36$, then value of $|\vec{n}|$ is equal to

35. $\hat{n}$ is a vector perpendicular to the plane containing the point whose position vectors are $\vec{a}, \vec{b}$ and $\vec{r}$, satisfying $\left|\left[\vec{r} - \vec{a} \ \vec{b} - \vec{a} \ \vec{n}\right]\right| = \lambda \left|\vec{r} \times \left(\vec{b} - \vec{a}\right) - \vec{a} \times \vec{b}\right|$ then the value of λ is

36. If $\vec{a}_1 = a\hat{i} + b\hat{j}, \vec{a}_2 = \vec{a}_1 - \left(2\hat{i} + \sqrt{5}\hat{j}\right)$ and $\vec{a}_3 = \vec{a}_1 + \left(2\hat{i} - \sqrt{5}\hat{j}\right)$, then

$$\begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$$ is equal to

37. If $\vec{a}$ and $\vec{b}$ are perpendicular vectors with $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{c} \times \vec{a} = \vec{b}$, then the least value of $|\vec{c} - \vec{a}|$

38. In a tetrahedron OABC, G is the centre of the tetrahedron, P, Q and R are the mid points of OA, OB and OC respectively. If the shortest distance between the line GR and PQ is $\frac{1}{2K}$, then the value of k is (where O is origin, $\overrightarrow{OA} = \hat{i}$, $\overrightarrow{OB} = \hat{j}$ and $\overrightarrow{OC} = \hat{k}$)

39. If $\vec{p}$ and $\vec{q}$ are orthogonal unit vectors and $\vec{s}$ is a vector such that $\vec{s} \times \vec{q} = \vec{p} - \vec{s}$ then value of $|\vec{s} + \vec{p} \times \vec{q}|$ is

40. Let a vector $\overline{V}_1 = \overline{OP}$, where O is origin and P is (2, 0, 2). Vector $\overline{V}_1$ is rotated about origin by an angle of $\frac{\pi}{3}$ in a plan which is perpendicular to XOZ plane to get vector $\overline{V}_2$ where $\overline{V}_2.\hat{j} > 0$. Again $\overline{V}_1$ is rotated about origin by right angle in XOZ plane to get vector $\overline{V}_3$. If V is volume of tetrahedron whose coterminous edges are $\overline{V}_1, \overline{V}_2$ and $\overline{V}_3$ then value of $\frac{3V}{\sqrt{6}}$ is ____

41. Let $\overline{A}, \overline{B}, \overline{C}$ be vectors of length 3, 4, 5 respectively. Let $\overline{A}$ be perpendicular to $\overline{B} + \overline{C}$, $\overline{B}$ is perpendicular to $\overline{C} + \overline{A}$ and $\overline{C}$ is perpendicular to $\overline{A} + \overline{B}$ and if the length of vector $\left|\overline{A} + \overline{B} + \overline{C}\right|$ is $k\sqrt{2}$ then find the value of k

42. Find the distance of the point $\hat{i} + 2\hat{j} + 3\hat{k}$ from the plane $\vec{r}.\left(\hat{i} + \hat{j} + \hat{k}\right) = 5$ measured parallel to the vector $2\hat{i} + 3\hat{j} - 6\hat{k}$.

43. The volume of a tetrahedron OABC formed by the vectors $\overline{OA} = \vec{a}, \overline{OB} = \vec{b}$ & $\overline{OC} = \vec{c}$ where $|\vec{a}| = 3, |\vec{b}| = 4, |\vec{c}| = 6$, angle between the vector – pairs $\{\vec{a}, \vec{b}\}; \{\vec{b}, \vec{c}\}$ and $\{\vec{c}, \vec{a}\}$ be $\frac{\pi}{3}$, $\cos^{-1}\frac{1}{4}$ and $\cos^{-1}\left(\frac{-1}{18}\right)$ respectively is V. If the value of V^2 can be written as $\frac{pq}{r^2}$ where p, q, r are prime numbers then evaluate $\frac{p+q}{r}$.

44. $|\overline{a}| = |\overline{b}| = |\overline{c}| = |\overline{a} + \overline{b}| = 1$. $\overline{a} \cdot \overline{c} = 0$

$\overline{a} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$, If for some $\overline{c}, |\overline{b} \times \overline{c}| = |\overline{a} \times \overline{c}|$ and $\overline{c} = p\hat{i} + q\hat{j} + r\hat{k}$,

Then value of $\left(\frac{p^4 + q^4 + r^4}{2}\right)$ is

(here $\vec{a}, \hat{k}$ and $\vec{b}$ are linearly dependent vectors).

45. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar unit vectors equally inclined to one another at an angle of $\frac{\pi}{3}$. If $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$ then $2\left(p^2 + q^2 + r^2\right)$ is equal to (where p, q, r are scalars)

46. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three unit vectors satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$, then $|3\vec{a} + 3\vec{b} + 8\vec{c}|$ equals

47. Let $\vec{a}$ and $\vec{b}$ be two mutually perpendicular vectors. Let $\vec{c} = x\vec{a} + y\vec{b} + \left(\vec{a} \times \vec{b}\right)$ for some x, y $\in$ R. If $|\vec{a}| = |\vec{b}| = 2$, $|\vec{c}| = 4$ and $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $\cos^2 \alpha$

48. Three lines are given by

$\vec{r} = \lambda\hat{i}, \ \lambda \in R$

$\vec{r} = \mu\left(\hat{i} + \hat{j}\right), \mu \in R$ and

$\vec{r} = v\left(\hat{i} + \hat{j} + \hat{k}\right), v \in R$.

Let the lines cut the plane $x + y + z = 1$ at the points A, B and C respectively. If the volume of a tetrahedron OABC is V (Where O is origin), then the value of 18V equals_______________

49. The plane $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$ intersect x-axis, y-axis, z-axis at A, B, C respectively. If the distance between origin and orthocentre of ΔABC is equal to k then the value of 7k is equal to ______

50. Let 'n' represents the number of Lattice points (x, y, z all being integers) inside the tetrahedron (not on the surface) having vertices $(0,0,0),(8,0,0),(0,8,0)$ and $(0,0,8)$ then the value of $\left(\frac{n}{7}\right)$ is

51. P, Q, R, S are the points $(1,2,-2),(8,10,11),(1,2,3)$ and $(3,5,7)$ respectively If s denotes the projection of PQ on RS then $29s^2+29$ is equal to $(8130-\lambda)$ where λ is________

52. The equation of the plane passing through the intersection of the planes $2x-5y+z=3$, $x+y+4z=5$ and parallel to the plane $x+3y+6z=1$ is $x+3y+6z=k$, where k is _________

53. Given $f(a,b,c)\le\frac{\pi}{2}$, where $f(a,b,c)=\left|\frac{a}{b}-\frac{1}{2}\right|+(3b-2c)^2+\sin^{-1}\left(1+\left((a-1)^2+(b-2)^2\right)\right)$. Also Π denotes the plane through the line $\frac{y}{b}+\frac{z}{c}=1$, x = 0 and parallel to the line $\frac{x}{a}-\frac{z}{c}=1$, y = 0. If the plane Π cuts the co-ordinate axes at A, B, C, then volume of tetrahedron OABC, O being the origin, is

54. If V be the volume of a tetrahedron and V' be the volume of the tetrahedron formed by joining the centroids of faces of given tetrahedron. If V = kV' then the value of k is

55. The plane ax + by + cz = 1 containing the line $\frac{x-3}{2}=\frac{y-1}{4}=\frac{z-2}{5}$ is rotated through an angle $\frac{\pi}{2}$ about this line to contain the point $(4,3,7)$. Value of $\left(2a+3b+\frac{c}{8}\right)$ equals

56. Given a, b $\in\{0,1,2,.......6\}$. Consider the system of equations

$x+y+z=4,$

$2x+y+3z=6,$

$x+2y+az=b$

If the number of ordered pairs (a, b) so that the system of equations has unique solution is λ, then $\frac{\lambda}{7}$ equals

57. If $\ell x+13y+mz+n=0$ is a plane through intersection of planes $2x+3y-z+1=0$ and $x+y-2z+3=0$ and is perpendicular to plane $3x-y-2z=4$, then $\frac{\ell-n}{m}$ equals

58. If the line $\frac{x-1}{15}=\frac{y+1}{16}=\frac{z-1}{n}$ intersect the curve $6x^2+5y^2=20, z=0$ then $9n^2+20n-k=0$, where k equals to ………

59. In the plane $x+y-2z=5$, P(a,b,c) is a point such that the sum of its distances from the point A (0, 1, -1) and B (3, 5, -1) is least then the value of (a + b + c) is

60. Let P be a variable point in the plane $6x+3y+2z-6=0$. Let A, B and C be the feet of perpendiculars drawn from P to planes

$P_1: x=0$

$P_2: y=0$ and

$P_3: z=0$ respectively. If locus of foot of perpendicular drawn from origin to plane passing through points A, B & C is $\left(x^2+y^2+z^2\right)\left(\frac{1}{x}+\frac{1}{2y}+\frac{1}{3z}\right)=n$, then the value of $\frac{1}{n}$ is

61. If <a, b, c> are direction cosines of line of intersection of three planes $x+y-z=1, (m-1)x+(3m-7)y-5z=-3$ and $(m-2)x+(2m-5)y=-4$ for some $m \in R$, then value of $\frac{a+b}{c}$ is

62. The equation of plane which passes through the line of intersection of the planes $x+y+z-1=0$ and $x-y+z-3=0$ and which is at maximum distance from the point (3, 4, 5) is $ax+by+cz=d$ then the value of $\frac{a+b-c}{4d}$ is

63. A straight line L intersects perpendicularly both the lines $\frac{x+2}{2}=\frac{y+6}{3}=\frac{z-34}{-10}$ and $\frac{x+6}{4}=\frac{y-7}{-3}=\frac{z-7}{-2}$ then the square of perpendicular distance of origin from L is equal to

64. Let OL, OM, ON be coterminous edges of a cuboid. OL = 1 cm, OM = 2 cm, ON = 3 cm. Let ℓ be the shortest distance between OL and skew body diagonal, m be the shortest distance between OM and skew body diagonal and n be the shortest distance between ON and skew body diagonal. Then the value of $\sqrt{13\ell^2+10m^2+5n^2}$ is

65. A plane is parallel to two lines whose direction ratios are (1, 0, -1) and (-1, 1, 2) and it contains the point (1, 1, 1). If it cuts coordinate axes at A, B and C, then the volume of the tetrahedron OABC is

66. Consider two straight lines $L_1 : \frac{x-2}{-\frac{1}{2}} = \frac{y-2}{\frac{1}{2}} = \frac{z-4}{1}$ and $L_1 : \frac{x-0}{\frac{1}{2}} = \frac{y+1}{1} = \frac{z-3}{-1}$. Let P and Q be the variable points on L_1 and L_2 respectively and point M divides PQ in the ratio 2 : 3 internally. If point M moves in a plane ax + bz = 6, then value of a + b is

Comprehension Type Questions

Paragraph for Questions 67&68

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three mutually perpendicular unit vectors

67. Let $\vec{r}$ be a unit vector satisfying $(\vec{b}-\vec{c})\times(\vec{r}\times\vec{a})+(\vec{c}-\vec{a})\times(\vec{r}\times\vec{b})+(\vec{a}-\vec{b})\times(\vec{r}\times\vec{c})=\vec{0}$. Then $\vec{r}$ is equal to

(A) $\frac{1}{\sqrt{3}}(\vec{a}+\vec{b}+\vec{c})$ only

(B) $-\frac{1}{\sqrt{3}}(\vec{a}+\vec{b}+\vec{c})$ only

(C) Both option (A) and option (B) are correct

(D) None of these

68. Let $\vec{r}$ be a vector having components 12,5 and 13 along $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. If λ is component of $\vec{r}$ along $(\vec{a}+\vec{b}+\vec{c})$, then $[\lambda]=$

(where [.] represents greatest integer function)

(A) 17 (B) 30 (C) 10 (D) 18

Paragraph questions from 69 to 71

A plane p contains the line $L_1 : \frac{y}{b}+\frac{z}{c}=1,\ x=0$ and is parallel to the line $L_2 : \frac{x}{a}-\frac{z}{c}=1, y=0$

69. Equation of plane p is

(A) $\frac{x}{a}-\frac{y}{b}+\frac{z}{c}=0$ (B) $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}-1=0$

(C) $\frac{x}{a}+\frac{y}{b}-\frac{z}{c}-1=0$ (D) $\frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$

70. If the shortest distance between L_1 and L_2 is $\frac{1}{4}$ then the value of $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$ equals

(A) 16 (B) 64 (C) 128 (D) 192

71. Distance of image $A(a,0,0)$ in the plane p from $M\left(-\frac{5}{3},\frac{8}{3},\frac{11}{3}\right)$, where $a=b=c=1$, is

(A) 1 (B) 2 (C) 3 (D) 4

Paragraph questions from 72 to 73

T is the region of the plane $x+y+6=1$ with $x,y,z>0$ is the set of points (a,b,c) in T such that just two of the following three inequalities hold: $a\le\frac{1}{2}, b\le\frac{1}{3}, c\le\frac{1}{6}$.

72. Area of the region T is

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) none of these

73. Area of the region S is

(A) $\frac{\sqrt{3}}{72}$ (B) $\frac{7\sqrt{3}}{36}$ (C) $\frac{\sqrt{3}}{4}$ (D) none of these

Paragraph questions from 74 to 75

Consider planes $P_1: x+2y+z-1=0$ & $P_2: x-y+2z+1=0$ and a plane P containing the line of intersection of P_1 & P_2.

74. If P is perpendicular to the plane $x+y+z=5$, then equation of P is :-

(A) $2x-4y+3z=3$ (B) $x+4y+3z=3$ (C) $x-4y+3z=0$ (D) $-x-4y+4z=0$

75. If from a point Q lying on a plane P in I^{st} octant the distance of planes $x=0, y=0$ & $z=0$ are 1, 1 and 2 respectively, then the vector equation of line passing through origin and parallel to normal of the plane P is

(A) $\vec{r}=\lambda\left(\hat{i}-14\hat{j}+2\hat{k}\right)$ (B) $\vec{r}=\lambda\left(2\hat{i}-\hat{j}+\hat{k}\right)$

(C) $\vec{r}=\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$ (D) $\vec{r}=\lambda\left(\hat{i}+14\hat{j}-3\hat{k}\right)$

(where $\lambda\in R$)

Paragraph questions from 76 to 77

Three equal spherical balls B_1, B_2, B_3 having centers at $\left(\frac{5}{\sqrt{3}},5,0\right),\left(\frac{5}{\sqrt{3}},-5,0\right)$ *and* $(\alpha,\beta,0)$ respectively touch other $\alpha>0$

76. Value of $(\alpha+\beta)$ is:

(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{5}{\sqrt{3}}$ (D) $\frac{20}{\sqrt{3}}$

77. Equation of common tangent plane at point of contact of $B_1 \& B_3$ is

(A) $15x+5\sqrt{3}y=225$ (B) $\sqrt{3}x-y=10$

(C) $5\sqrt{3}x-15y=225\sqrt{3}$ (D) $2\sqrt{3}x+2y+25=0$

Paragraph questions from 78 to 79

Let $L_1:\vec{r}=2\hat{i}+\hat{j}-\hat{k}+\lambda\left(\hat{i}-2\hat{j}+\hat{k}\right)$ and $L_2:\vec{r}=\frac{8}{3}\hat{i}-3\hat{j}-\hat{k}+\mu\left(2\hat{i}-\hat{j}+\hat{k}\right)$ where $\lambda,\mu\in R$ be two lines in space. A line L passing through origin meet the lines L_1 and L_2 at P and Q respectively.

78. $M(1,2,3)$ be a point in space and N be a point on the lines , then the value of λ for which $\overrightarrow{MN}$ is parallel to the plane $\vec{r}.\left(\hat{i}-4\hat{j}+3\hat{k}\right)=1$, is equal to

(A) $\frac{-3}{4}$ (B) $\frac{5}{8}$ (C) $\frac{-2}{9}$ (D) $\frac{7}{12}$

79. The equation of the plane passing through the points P and Q and perpendicular to the plane $\vec{r}.\left(\hat{i}-\hat{j}+\hat{k}\right)+1=0$ is

(A) $\vec{r}.\left(2\hat{i}-\hat{j}\right)=0$ (B) $\vec{r}.\left(3\hat{i}+2\hat{j}\right)=0$

(C) $\vec{r}.\left(\hat{i}+\hat{j}\right)=0$ (D) $\vec{r}.\left(\hat{i}-\hat{j}\right)=0$

MATCHING LIST TYPE QUESTIONS

80. List-I contains vector equation in unknown $\vec{r}$ and List -II contains number of solutions i.e. number of such possible $\vec{r}$.

List-I	List-II
(P) $\vec{r}\times\hat{i}=\hat{i}$	(1) 0
(Q) $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\left\|\vec{r}\right\|=\frac{1}{\sqrt{3}}$	(2) 1
(R) $\left\|\vec{r}-\hat{i}\right\|=1$	(3) more than 1

(A) (P) – (1); (Q) – (2); (R) – (3) (B) (P) – (2); (Q) – (1); (R) – (3)

(C) (P) – (3); (Q) – (2); (R) – (1) (D) (P) – (1); (Q) – (3); (R) – (2)

Matrix Matching Type Questions:

81. **Match the following statement in List-I with the values in List-II**

List-I	List-II
(p) The projection of the vector $\left(3\hat{i}-4\hat{j}+2\hat{k}\right)$ on the line segment Joining the point (2, 3, -1) and (-2, -4, 3) is	(1) $\frac{1}{2}$
(Q) Let $\bar{\alpha},\bar{\beta},\bar{\gamma}$ are three vectors such that $\bar{\alpha}+\bar{\beta}+\bar{\gamma}=0$ and $\lvert\bar{\alpha}\rvert=3,\lvert\bar{\beta}\rvert=5,\lvert\bar{\gamma}\rvert=7$. If θ is the angle between $\bar{\alpha}$ and $\bar{\beta}$ then value of $\cos\theta$ is	(2) 3
(R) If $\bar{A}=3\hat{i}-\hat{j}-2\hat{k}$ and $\bar{B}=2\hat{i}+3\hat{j}+\hat{k}$, then the value of $\left\lvert\left(\bar{A}+\bar{B}\right)\times\left(\bar{A}-\bar{B}\right)\right\rvert$ is $2\sqrt{P}$. Value of $\frac{P}{65}$ is	(3) $\frac{8}{3}$
(S) If the unit vectors $\bar{a},\bar{b},\bar{c}$ are such that $\bar{a}+\bar{b}+\bar{c}=\bar{0}$ Then the value of $\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a}$ is	(4) $-\frac{3}{2}$

(A) $P\to3;Q\to1;R\to4;S\to2$ (B) $P\to3;Q\to4;R\to1;S\to2$

(C) $P\to1;Q\to2;R\to3;S\to4$ (D) $P\to3;Q\to1;R\to2;S\to4$

Matrix Matching Type Questions:

Answer Q.82, Q.83 and Q.84 by appropriately matching the information given in the three columns of the following table.

$\vec{a}$ and $\vec{b}$ are unit vectors inclined at an angle of $60°$ and $\vec{r}=\lambda\vec{a}+\mu\vec{b}+\gamma\left(\vec{a}\times\vec{b}\right)$ Match conditions given in **Column-I** with values of $\lvert\lambda-\mu\rvert$ in **Column-II** and $\lvert\gamma\rvert$ in **Column-III**

Column-I	Column-II	Column-III
(I) $\vec{r}.\vec{a}=0,\vec{r}.\vec{b}=2,\left(\vec{r}\times\vec{a}\right).\vec{b}=1$	(i) $\frac{4}{3}$	(P) 0
(II) $\vec{r}.\vec{a}=0,\vec{r}.\vec{b}=1$ and $\vec{r}$ lies in plane of $\vec{a}$ and $\vec{b}$	(ii) 2	(Q) $\frac{2}{3}$
(III) $\vec{r}\times\vec{a}=3\vec{a}\times\vec{b},\vec{r}.\vec{a}=1$	(iii) 4	(R) $\frac{4}{3}$

(IV) $\vec{a}\times(\vec{b}\times\vec{r})=\vec{a}\times\vec{b}, \vec{a}.\vec{r}=2$ (iv) $\frac{11}{2}$ (S) 2

82. Which of the following combination is **CORRECT**?

(A) (I) (i) (R) (B) (I) (i) (Q) (C) (II) (ii) (s) (D) (II) (ii) (P)

83. The CORRECT Combination of (III) of Column-I is

(A) (ii) (P) (B) (i) (S) (C) (iii) (S) (D) (iv) (P)

84. The CORRECT Combination with (iii) of Column-II is

(A) (I) (Q) (B) (II) (R) (C) (III) (P) (D) (IV) (S)

Match the following

List-I		**List-II**	
(I)	If $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}-\hat{j}+\hat{k}, \vec{c}=\hat{i}+2\hat{j}-\hat{k}$, then The value of $\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c}\end{vmatrix}$ is divisible by	(P)	(6)
(II)	If points P, Q and R have position vectors, $\vec{r_1}=3\hat{i}-2\hat{j}-\hat{k}, \vec{r_2}=\hat{i}+3\hat{j}+4\hat{k}$ and $\vec{r_3}=2\hat{i}+\hat{j}-2\hat{k}$ respectively, relative to an origin O, then the value of distance of P from the plane OQR, divides………	(Q)	28
(III)	A line with direction ratios (2, 1, 2) intersects the lines, $\vec{r}=-\hat{j}+\lambda(\hat{i}+\hat{j}+\hat{k})$ and $\vec{r}=-\hat{i}+\mu(2\hat{i}+\hat{j}+\hat{k})$ at A and B, then the value of length (AB) divides………	(R)	16
(IV)	In a regular tetrahedron, the centres of the four faces are the vertices of a smaller tetrahedron. The ratio of the volume of the smaller tetrahedron to that larger is $\frac{m}{n}$, where m and n are relatively prime positive integers. The value of (m + n) is	(S)	3
		(T)	8
		(U)	10

85. Which of the following is the only correct combination?

(A) I $\rightarrow$ R, T (B) II $\rightarrow$ R, T (C) I $\rightarrow$ Q, T (D) II $\rightarrow$ P, T

86. Which of the following is the only correct combination?

(A) III $\rightarrow$R, T (B) IV $\rightarrow$ T (C) III $\rightarrow$ P,S (D) IV $\rightarrow$ R

87. Consider line as shown $L_1 : y = ax, z = c; L_2 : y = -ax, z = -c, ax = -c; L_3 : y = z, ax = -c$. Then L is a variable line which intersects the given lines $(a \neq b \neq c)$ match the following Column-I with Column-II.

Column-I	Column-II
(A) Locus of L passes through point P where P is	p) $(a^2, 1, -c^2)$
(B) Locus of L is $k_1x^2 - y^2 + k_2z^2 + k_3 = 0$ (k_1, k_2, k_3) is	q) $(0, 0, c)$
(C) Locus L intersects the xz-plane in Locus L' which is an ellipse with major axis α ,minor axis β ,centre as (α_1, β_1) where $(\alpha, \beta, \alpha_1 + \beta_1)$ $(a^2 > 1)$	r) $\left(\frac{2c}{a}, 1, 0\right)$
(D) Locus L intersects the xy-plane in hyperbola with transeverse axis α' conjugate is β', centre as (α'_1, β'_1) where $(\alpha', \beta', \alpha'_1, \beta'_1)$ is $\left(\frac{c^2}{a^2} > 1\right)$	s) $\left(2c, \frac{2c}{a}, 0\right)$
	t) $\left(\frac{c}{a}, 0, 0\right)$

(A) (A) – q,t; (B) – p; (C) – s; (D) – r (B) (A) – q,s; (B) – r; (C) – q; (D) – t

(C) (A) – q,s; (B) – s; (C) – p; (D) – q (D) (A) – q,t; (B) – t; (C) – p; (D) – s

88. Match the two columns.

A plane π passes through the point $(1,1,1)$ and is parallel to the vectors $\vec{b}=(1,0,-1)$ and $\vec{a}=(-1,1,0)\pi$ meets the axes in A, B and C forming the tetrahedron OABC.

Column I	**Column II**
(A) Volume of the tetrahedraon OABC in cubic units.	(p) $\frac{9}{2}$
(B) Area of a face of the tetrahedron in square units.	(q) $\frac{9\sqrt{3}}{2}$
(C) Distance of the points, $\left(\frac{3\sqrt{3}}{2}, 3\sqrt{3}, 3\right)$ from the plane π in units.	(r) $\sqrt{3}$
(D) Distance of a face of the tetrahedron from the vertex	(s) 3

(A) $P-1, Q-1, 2.R-1, S-4$ (B) $P-2, Q-3, R-2, S-3$

(C) $P-3, Q-3, R-2, S-2$ (D) $P-4, Q-4, 2.R-3, S-3$

Linked Comprehension Type Questions

Paragraph for Questions 89&90

Linked comprehensive type passage:

Equation of line L = 0 is given by $2x-y+z-2=0=x+2y-z-3$, then

89. The equation of the plane containing the line L = 0 and passing through the point (3, 2, 1) is $ax+by+cz+1=0$, then $a-b+c$ is divisible by

(A) 3 (B) 4 (C) 2 (D) 6

90. Vector equation of the plane which passes through the point (-1, 3, 2) and is perpendicular to line L = 0 is

(A) $\vec{r}.\left(\hat{i}-3\hat{j}-5\hat{k}\right)+20=0$ (B) $\vec{r}.\left(\hat{i}-3\hat{j}+5\hat{k}\right)=0$

(C) $\vec{r}.\left(\hat{i}+3\hat{j}+5\hat{k}\right)-18=0$ (D) $\vec{r}.\left(-\hat{i}+3\hat{j}-5\hat{k}\right)+3=0$

Paragraph for Questions 91 & 92

L_1 and L_2 are two lines given by the equations $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1} : \frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$ respectively then

91. Shortest distance between the lines is

(A) $\sqrt{29}$ (B) $2\sqrt{29}$ (C) 6 (D) $2\sqrt{31}$

92. The equation of line of shortest distance between the lines is

(A) $\frac{x-3}{2}=\frac{y-5}{3}=\frac{z-7}{4}$ (B) $\frac{x+3}{2}=\frac{y+5}{3}=\frac{z-7}{4}$

(C) $\frac{x+3}{2}=\frac{y-5}{3}=\frac{z-7}{4}$ (D) $\frac{x-3}{2}=\frac{y-5}{3}=\frac{z+7}{4}$

Answer the following by appropriately matching the lists based on the information given in the paragraph.

$P(\alpha_1,\alpha_2,\alpha_3)$ is the image of (2, 1, 4) in the plane $2x-y+z+5=0$ and $Q(\beta_1,\beta_2,\beta_3)$ is the foot of perpendicular from (2, 1, 4) in the plane $2x-y+z+5=0$, then

List-I	List-II
(I) $\alpha_1+\alpha_2+\alpha_3$	(P) -1
(II) $\alpha_2+\alpha_3$	(Q) 5
(III) $\alpha_1+\beta_1$	(R) 3
(IV) $\beta_1+\beta_2$	(S) -8
	(T) 1

93. Which of the following is correct?

(A) I-P, III-R (B) I-P, II-T (C) II-R, IV-S (D) I-P, II-Q, III-S

Answer the following by appropriately matching the lists based on the information given in the paragraph

Consider the planes

$P_1: x+y-z=5$

$P_2: 2x-y+\lambda z=3$ (where $\lambda\in R$)

$P_3: x-3y-4=0$

$P_4: 4y-z+5=0$

Plane P_1 and P_2 intersect to form line L_1. Plane P_3 and P_4 intersect to form line L_2. There are some expression given in the List-I match them with the List-II given below

List-I	**List-II**
(I) Value of λ for which L_1 and L_2 are coplanar	(P) $-\frac{5}{4}$
(II) Image of the line L_2 in the plane P_1 is passing Through	(Q) (0, -33, 0)
(III) Image of plane P_1 in the plane P_3 is passing through	(R) $-\frac{6}{5}$
(IV) Equation of the line L_2 is passing through	(S) (1, -1, 1)
	(T) (-1, 1, -11)
	(U) (11, 5, -5)

94. Which of the following is the only CORRECT combination?

(A) (I), (P) (B) (II), (S) (C) (III), (U) (D) (IV), (T)

95. Which of the following is the only INCORRECT combination?

(I), (P) (B) (II), (S) (C) (III), (Q) (D) (IV), (S)

THREE COLUMN MATCHING

Answer questions 96, 97 and 98 by appropriately matching the information given in the three columns of the following table. Consider three plans:

$P_1 = 2x + y + z = 1$

$P_2 = x - y + z = 2$

$P_3 = \alpha x - y + 3z = 5$

The three planes intersects each other at point P on XOY plane and at point Q on YOZ plane. O is the origin.

Column 1	**Column 2**	**Column 3**
(I) The value of α is	(i) 1	(P) root of equation $x^2 - x(4+i) + 4i = 0$
(II) The length of projection of PQ on x – axis is	(ii) 2	(Q) root of equation $x^2 - 1 = 0$
(III) If the coordinates of point R situated at minimum distance from point O on the line PQ are (a, b, c), then the value of $7a + 14b + 14c$ is	(iii) 4	(R) root of equation $x^3 - 8 = 0$

(IV) If the area of ΔPOQ is $\sqrt{\frac{a}{b}}$. Then the value of a – b is where a and b are co-prime numbers	(iv) 3	(S) root of equation $x^2+2x-15=0$

96. Which is correct combination?

(A) (I) (i) (P) (B) (I) (iii) (P) (C) (II) (ii) (Q) (D) (III) (ii) (Q)

97. Which is correct combination?

(A) (I) (ii) (Q) (B) (II) (i) (Q) (C) (III) (iii) (R) (D) (IV) (iv) (Q

98. Which is correct combination?

(A) (I) (ii) (Q) (B) (III) (ii) (R) (C) (IV) (iv) (R) (D) (I) (ii) (R)

STEM BASED QUESTIONS

STEM-I

Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-coplanar unit vector such that the angle between every pair of them is $\frac{\pi}{3}$. If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c},$ where p, q, r are scalars; then

99. The value of $\left[\vec{a}\vec{b}\vec{c}\right]^4$ is ______

100. The value of $=20p^2+21q^2-22r^2$ is ________

STEM - II

Where $\gamma\in R$ and $\theta\in\left[0,2\pi\right]$ and min value of determinant M is 8 and foot of perpendicular from point $P\left(\alpha-4,\beta-10,1\right)$(where $\left(\alpha,\beta\right)$ for which the above system of linear equation has infinitely many solution) to the plane $x+2y-2z-10=0$ is F (a, b, c), then

101. The value of $\frac{\gamma}{5}$ is _______

102. The value of $\frac{3}{2}\left(a+b+c\right)$ is _________

KEY

1	**A**	2	**A**	3	**A**	4	**A**	5	**C**
6	**C**	7	**D**	8	**A**	9	**B**	10	**C**
11	**B**	12	**A**	13	**AC**	14	**ABCD**	15	**ACD**
16	**BD**	17	**AC**	18	**BD**	19	**ABCD**	20	**AD**
21	**ABC**	22	**BC**	23	**AC**	24	**BC**	25	**AD**
26	**AD**	27	**ACD**	28	**ACD**	29	**BC**	30	**ACD**
31	**ABC**	32	**BC**	33	**ABC**	34	**4**	35	**1**
36	**80**	37	**1.50**	38	**2.44**	39	**0.70**	40	**4**
41	**5**	42	**7**	43	**30**	44	**0.25**	45	**3**
46	**5**	47	**0**	48	**0.50**	49	**6**	50	**5**
51	**1**	52	**7**	53	**1**	54	**27**	55	**7.75**
56	**6**	57	**4**	58	**2630**	59	**2**	60	**0.50**
61	**1**	62	**1.25**	63	**5**	64	**7**	65	**0.16**
66	**3**	67	**C**	68	**A**	69	**D**	70	**B**
71	**C**	72	**B**	73	**B**	74	**C**	75	**D**
76	**D**	77	**B**	78	**D**	79	**C**	80	**A**
81	**D**	82	**D**	83	**D**	84	**D**	85	**A**
86	**C**	87	**A**	88	**A**	89	**ACD**	90	**A**
91	**B**	92	**A**	93	**D**	94	**A**	95	**B**
96	**B**	97	**B**	98	**B**	99	**0.25**	100	**9.50**
101	**0.20**	102	**2.50**						

SOLUTIONS

1. $\overline{d} = x\overline{a} + y\overline{b} + z\overline{c}$ take dot product with

$\overline{b} \times \overline{c}, \overline{c} \times \overline{a}, \overline{a} \times \overline{b}$

$$\overline{d} = \frac{\left[\overline{d}\,\overline{b}\,\overline{c}\right]a + \left[\overline{d}\,\overline{c}\,\overline{a}\right]\overline{b} + \left[\overline{d}\,\overline{a}\,\overline{b}\right]\overline{c}}{\left[\overline{a}\,\overline{b}\,\overline{c}\right]}$$

2. $\overrightarrow{GA} = \overrightarrow{2b} - \vec{a}$

$\overrightarrow{CA} = \vec{b} - \vec{a}$

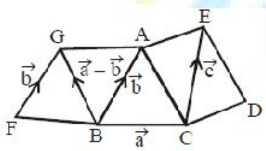
$\overrightarrow{AE} = \overrightarrow{AC} + \overrightarrow{CE} = \vec{a} - \vec{b} + \vec{c}$

$\overrightarrow{GE} = \overrightarrow{GA} + \overrightarrow{AE}$

$= 2\vec{b} - \vec{a} + \vec{a} - \vec{b} + \vec{c}$

$= \vec{b} + \vec{c}$

$\Rightarrow \therefore \alpha = 0,\ \beta = 1, \gamma = 1$



3. Here $\vec{d} = \dfrac{\vec{a} + \vec{b} + \vec{c}}{\sqrt{3}}$

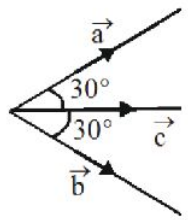
$\left|\vec{a} + \vec{b} + \vec{c} + \vec{d}\right|^2 = \left|\vec{a} + \vec{b} + \vec{c}\right|^2 + \left|\vec{d}\right|^2 + 2\vec{d}\left(\vec{a} + \vec{b} + \vec{c}\right)$

$= 4 + 2\sqrt{3}$

4. $V^2 = \dfrac{1}{36}\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2 = \dfrac{1}{36}\begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} = \dfrac{1}{12}\sqrt{3\sqrt{3} - 5}$

5. $a^2 + b^2 + c^2 + 2\vec{a}.\vec{b} + 2\vec{b}.\vec{c} + 2\vec{c}.\vec{a}$

$= 14 + 2 \times \left(2 \times \dfrac{1}{2} + 6 \times \dfrac{\sqrt{3}}{2} + \dfrac{3\sqrt{3}}{2}\right)$



$= 14 + 2 + 6\sqrt{3} + 3\sqrt{3} = 16 + 9\sqrt{3}$

6. $\left[\vec{u}\,\vec{v}\,\vec{w}\right] = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} \left[\vec{a}\ \vec{b}\ \vec{c}\right]$

$= \left[\vec{a}\,\vec{b}\,\vec{c}\right]^3$

$\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \begin{vmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & 2 & 3 \end{vmatrix} = \left(\sqrt{3} - \frac{1}{2}\right)$

7. Function |t| is not differentiable only at t = 0, whereas function $\left|\sin \pi t\right|$ is differentiable everywhere except at all integral points. Also, function $\sqrt[3]{t-2}$ is not differentiable only at t = 2. So, $\vec{r}(t)$ is differentiable everywhere except at all integral points.

8. $\begin{cases} (2x + y - 1) + \lambda(x - 2y + 3z) = 0 \\ (3x - y + z + 2) + \mu(4x + 5y - 2z - 3) = 0 \end{cases}$

Line is parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$

So, $(2+\lambda)\times 1 + (1-2\lambda)\times 2 + 3\lambda \times 3 = 0$

$\lambda = -\frac{2}{3}$

$(3+\mu)\times 1 + (-1+5\mu)\times 3 = 0$

$\mu = -\frac{1}{2}$

9. Equation of plane containing L_1 is

$P_1 + \lambda P_1 = 0 \qquad \ldots(1)$

Equation of plane containing L_2 is

$P_3 + \mu P_4 = 0 \qquad \ldots.(2)$

For some λ and μ equation (1) and (2)

Are representing the same plane

$\Rightarrow$ equation of plane containing L_1 and L_2 is

$-7x + 7y - 8z - 3 = 0$

Let $Q(\lambda, \lambda, \lambda)$ be a point in the plane

$\Rightarrow \lambda = -\frac{3}{8}$

so, required distance $= \sqrt{3}|\lambda| = \frac{3\sqrt{3}}{8}$

10. Let $P(\lambda, \lambda - 2, \lambda)$ and a $\left(\mu - 2, \frac{\mu}{2}, \frac{\mu}{2}\right)$

Then $\frac{\lambda - \mu + 2}{2} = \frac{\lambda - 2 - \frac{\mu}{2}}{1} = \frac{\lambda - \frac{\mu}{2}}{2}$

$\Rightarrow \qquad \lambda - \mu + 2 = 2\lambda - \mu - 4$

$\lambda - \mu + 2 = \lambda - \frac{\mu}{2}$

$\Rightarrow \mu = 4, \lambda = 6$

P(6, 4, 6) Q(2, 2, 2)

$PQ = \sqrt{16 + 4 + 16} = 6$

11. $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-3}{1} = \frac{-2 \times 3}{3}$

$\Rightarrow P(-1, 4, 1), Q(3, 1, 7)$

$$\vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & 6 \\ 1 & 2 & 1 \end{vmatrix} = -15\hat{i} + 2\hat{j} + 11\hat{k}$$

Required plane is : $15x - 2y - 11z + 34 = 0$

12. Let $(\lambda, \lambda, \lambda)$ point of intersection then $\lambda(\sin A + \sin B + \sin C) = 2d^2$ and $\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$

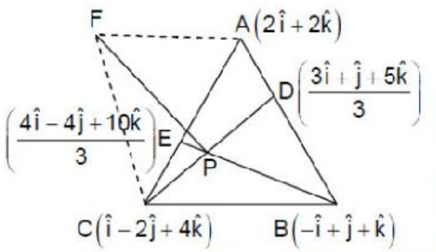
dividing $\frac{\sum \sin 2A}{\sum \sin A} = \frac{1}{2}$

$\Rightarrow \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=\frac{1}{16}$

13. $CD:\vec{r}=\left(\hat{i}-2\hat{j}+4\hat{k}\right)+\frac{\lambda}{3}\left(7\hat{j}-7\hat{k}\right)$

$BE:\vec{r}=\left(-\hat{i}+\hat{j}+\hat{k}\right)+\frac{\mu}{3}\left(7\hat{i}-7\hat{j}+7\hat{k}\right)$

$P=\left(\hat{i}-\hat{j}+3\hat{k}\right)$



Area of tetrahedron $ABCF$

$=\frac{1}{3}(Area\,of\,base\,triangle)\times height=\frac{7}{3}cubic\,units$

$\overrightarrow{AB}\times\overrightarrow{AC}=7\hat{i}+7\hat{k}, \left|\overrightarrow{PF}\right|=PF=\sqrt{2}units$

$\overrightarrow{PF}=\sqrt{2}\left(\frac{7\hat{j}+7\hat{k}}{\sqrt{49+49}}\right)=\hat{j}+\hat{k}=P.V\ of\ F-P.V.of\ P$

$P.V.of\ F=\hat{i}+4\hat{k}$

Vector equation of AF is $\vec{r}=2\left(\hat{i}+\hat{k}\right)+a\left(-\hat{i}+2\hat{k}\right)$

14. By eliminating n $2\left(\frac{\ell}{m}\right)^2-\frac{\ell}{m}-1=0$

$\Rightarrow \frac{\ell_1}{m_1}=1, \frac{\ell_2}{m_2}=-\frac{1}{2}$

On solving we $\left(\ell_1,m_1,n_1\right)=\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},-\sqrt{\frac{2}{3}}\right) and \left(\ell_2,m_2,n_2\right)=\left(\frac{1}{\sqrt{6}},-\sqrt{\frac{2}{3}},\frac{1}{\sqrt{6}}\right)$

15. (A) Intersection point is mid point of AB and CD

(B) $2\alpha+6\beta+15\gamma=7$

(C) Triangle is Equilateral

$\left(2\hat{i}+6\hat{j}+3\hat{k}\right).\left(\alpha\hat{i}+\beta\hat{j}+5\gamma\hat{k}\right)\le 7\times\sqrt{\alpha^2+\beta^2+25\gamma^2}$

$$\left(2\alpha+6\beta+15\gamma\right)\le 7\times\sqrt{\alpha^2+\beta^2+25\gamma^2}$$

$$\alpha^2+\beta^2+25\gamma^2\ge 1$$

(D) Intersection points are $(1,2,3)$ and $(2,-3,1)$

16. $\vec{a}\times\vec{b}=x\vec{a}+y\vec{b}+z\vec{c}$. Let $\vec{a}=\lambda\vec{b}\times\vec{c}.\lambda=\dfrac{1}{2}$

So, $\dfrac{\vec{b}\times\vec{c}}{2}\times\vec{b}=\dfrac{x}{2}\vec{b}\times\vec{c}+y\vec{b}+z\vec{c}$

$$\Rightarrow -\frac{\vec{b}}{2}+\frac{\vec{c}}{6}=\frac{x}{2}\left(\vec{b}\times\vec{c}\right)+y\vec{b}+z\vec{c}$$

$$\Rightarrow x=0, y=-\frac{1}{2}, z=\frac{1}{6}$$

17. Any vector r in space can be written as

$$\vec{r}=\lambda\vec{a}+\mu\vec{b}+k\left(\vec{a}\times\vec{b}\right)$$

Take dot product with $\vec{a}\times\vec{b}$, we get $k=\dfrac{5}{4}$

Let $\vec{r}.\vec{a}=\lambda;\vec{r}.\vec{b}=\mu$

$$\because r.\left(3\vec{a}+2\vec{b}\right)=0\Rightarrow 3\lambda+2\mu=0$$

Given $\displaystyle\int_{-2\lambda}^{-\frac{4}{3}\mu}\frac{x+1}{x^2+1}dx=\int_{-2\lambda}^{2\lambda}\frac{x+1}{x^2+1}dx$

$$=2\int_0^{2\lambda}\frac{dx}{x^2+1}=2\tan^{-1}(2\lambda)=\frac{\pi}{2}$$

$$\Rightarrow \lambda=\frac{1}{2}\text{ and }\mu=-\frac{3}{4}$$

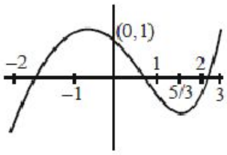
$$\therefore \vec{r}=\frac{\vec{a}}{2}-\frac{3}{4}\vec{b}+\frac{5}{4}\left(\vec{a}\times\vec{b}\right)$$

And $|\vec{r}|=\sqrt{\dfrac{1}{4}+\dfrac{9}{16}+\dfrac{25}{16}}\Rightarrow|\vec{r}|^2=\dfrac{19}{8}$]

18. Clearly, $\begin{vmatrix} x & x & 5 \\ 1 & x & 0 \\ 1 & 1/5 & x \end{vmatrix}=0\Rightarrow x^3-x^2-5x+1=0$

Assume $f(x) = x^3 - x^2 - 5x + 1$

$\therefore f(x) = 3x^2 - 2x - 5 = 0 \Rightarrow x = -1$ & $\frac{5}{3}$



Clearly $[\alpha]+[\beta]+[\gamma] = -2+0+2 = 0$

Now $x^3 - x^2 - 5x + 1 = (x-\alpha)(x-\beta)(x-\gamma)$

Put $x = -1$, we get

$(1+\alpha)(1+\beta)(1+\gamma)$

$= -(-1-1+5+1) = -4$

Now, volume of parallelised

$= \left|x^3 - x^2 - 5x + 1\right| [x \in -2, 3]$

Maximum value will occur at $x = \frac{5}{3}$ is equal to

$$\left|\frac{125}{27} - \frac{25}{9} - \frac{25}{3} + 1\right| = \left|\frac{125 - 75 - 225 + 27}{27}\right| = \frac{148}{27}$$

19. $x\vec{a} + y\vec{b} - \vec{c} = \vec{0} \Rightarrow$ if $x + y - 1 = 0$ then C lies on line AB.

If $x > 0$, $y > 0$ and $x + y - 1 = 0$ then C lies on line segment AB.

If $x < 0$, $y > 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment AB.

If $x > 0$, $y < 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment AB.

f $x > 0$, $y < 0$ and $x + y - 1 < 0$ then C lies inside ΔAOB.

20. As $\vec{\alpha}, \vec{\beta}$ are non collinear unit vectors

$\Rightarrow \vec{\alpha}, \vec{\beta}, \vec{\alpha} \times \vec{\beta}$ are non-coplanar non-zero vectors

$\Rightarrow a - b = b - c = c - a = 0$

21. $\left[\vec{a}\times\vec{p}\ \vec{b}\times\vec{q}\ \vec{c}\times\vec{r}\right] + \left[\vec{a}\times\vec{q}\ \vec{b}\times\vec{r}\ \vec{c}\times\vec{p}\right] + \left[\vec{a}\times\vec{r}\ \vec{b}\times\vec{p}\ \vec{c}\times\vec{q}\right]$

$\left[\vec{a}\times\vec{p}\ \vec{b}\times\vec{q}\ \vec{c}\times\vec{r}\right]$

$= (\vec{a}\times\vec{p}).\left\{(\vec{b}\times\vec{q})\times(\vec{c}\times\vec{p})\right\}$

$= \{\vec{a}\times\vec{p}\}.\left\{\left[\vec{b}\,\vec{q}\,\vec{r}\right]\vec{c} - \left[\vec{b}\,\vec{q}\,\vec{c}\right]\vec{r}\right\}$

$\left[\vec{a}\times\vec{p}\ \vec{b}\times\vec{q}\ \vec{c}\times\vec{r}\right]$

$= \left[\vec{a}\ \vec{p}\ \vec{c}\right]\left[\vec{b}\ \vec{q}\ \vec{r}\right] - \left[\vec{a}\,\vec{p}\,\vec{r}\right]\left[\vec{b}\ \vec{q}\ \vec{c}\right]$ .... (i)

$\left[\overline{a}\times\overline{q}\ \overline{b}\times\overline{r}\ \overline{c}\times\overline{p}\right]$

$= \left\{(\overline{a}\times\overline{q})\times(\overline{b}\times\overline{r})\right\}.(\overline{c}\times\overline{p})$

$= \left[\overline{b}\,\overline{r}\,\overline{a}\right]\left[\overline{q}\,\overline{c}\,\overline{p}\right] - \left[\overline{b}\,\overline{r}\,\overline{q}\right]\left[\overline{a}\,\overline{c}\,\overline{p}\right]$ ..... (ii)

$\left[\overline{a}\times\overline{r}\,\overline{b}\times\overline{p}\,\overline{c}\times\overline{q}\right]$

$= (\overline{a}\times\overline{r}).\left\{(\overline{b}\times\overline{p})\times(\overline{c}\times\overline{q})\right\}$

$= \left[\overline{c}\,\overline{q}\,\overline{b}\right]\left[\overline{a}\,\overline{r}\,\overline{p}\right] - \left[\overline{a}\,\overline{r}\,\overline{b}\right]\left[\overline{c}\,\overline{q}\,\overline{p}\right]$ ..... (iii)

Now (i) + (ii) + (iii)

22. $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 0$

So $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors

As $\vec{p}$ is equally inclined to $\vec{a}, \vec{b}, \vec{c}$.

$\Rightarrow \vec{p}$ is perpendicular to $\vec{a}, \vec{b}, \vec{c}$.

$\therefore \vec{p}.\vec{a} = \vec{p}.\vec{b} = \vec{p}.\vec{c} = 0$

$\therefore \vec{p}.\vec{x} = \vec{p}.\vec{y} = \vec{p}.\vec{z} = 0$

$\vec{p} = \lambda\left(\vec{a}\times\vec{b}\right) = \lambda\left(-\hat{i}+2\hat{j}+3\hat{k}\right)$

$\vec{p}.\left(2\hat{i}-3\hat{j}+4\hat{k}\right) = \lambda(-2-6-12) \neq 0$

24. Let the direction ratios of the two lines be $(0, m_1, n_1)$ and (l_2, m_2, n_2) so that $2m_1 - n_1 = 0$,

$l_2 + 2m_2 - n_2 = 0$ and $m_1m_2 + n_1n_2 = 0$

$\Rightarrow \frac{l_1}{0} = \frac{m_1}{1} = \frac{n_1}{2} \Rightarrow \frac{l_2}{5} = \frac{m_2}{-2} = \frac{n_2}{1}$.

Hence equation of the required line are

$\frac{x-1}{5}=\frac{y-2}{-2}=\frac{z+1}{1}=1$.

For $\lambda=-1$ *and* $\lambda=1$ we get (B) *and* (C)

25. Line L_2 lie in a plane P

$\therefore 2\alpha-\alpha+7=0 \Rightarrow \alpha=-7$

$\& -2\alpha+2-\beta=0 \Rightarrow \beta=16$

line L_1 is parallel to Plane P

$\therefore$ Minimum distance is $\frac{23}{3}$

26. $\begin{vmatrix} x & y & z \\ 2 & -3 & 1 \\ 1 & 1 & 1 \end{vmatrix}=0; \begin{vmatrix} x-2 & y-1 & z+2 \\ 3 & -5 & 2 \\ 1 & 1 & 1 \end{vmatrix}=0$

27. Lines L_1 and L_2 are intersects at point (1, 0, 2),

L_1, L_2 and $\vec{r}=4\hat{j}+\lambda(\hat{i}+2\hat{k})$ are always coplanar $\forall \lambda \in R$.

Any line passes through point $A(1,\mu,3)$ and intersecting L_1 and L_2 will always passes through (1, 0, 2)

28. $\Delta=\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix}=0$ (for Non trivial solution)

$-(p+q+r)(p^2+q^2+r^2-pq-qr-rp)=0$

$p+q+r=0$

Represents line $x=y=z$

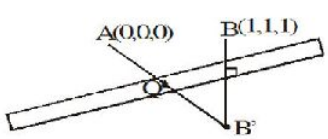
$p^2+q^2+r^2-pq-qr-rp=0$

$\frac{1}{2}(p-q)^2+(q-r)^2+(r-p)^2=0$

P, q, r must be equal $p=q=r$

$\Rightarrow$ identical planes $x+y+z=0$

29.



For (QA + QB) min, Q must lie on AB' line

Where B' is mirror image of B in plane $x - y + z = 3$

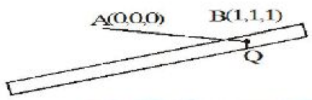
$B' \equiv \left(\frac{7}{3}, \frac{-1}{3}, \frac{7}{3}\right)$

Q is intersection of AB' and plane

$Q\left(\frac{7}{5}, \frac{-1}{5}, \frac{7}{5}\right) \equiv (a, b, c)$

Hence a – c- 10 b = 2

For $|QA - QB|_{min}$, Q must lie on point of intersection of AB with plane



$Q \equiv (3,3,3) \equiv (a,b,c)$

Hence $a - 2b + c = 0$

30. Any point on the given line will be

$(2\lambda + 1, -3\lambda - 1, \lambda) \equiv (\alpha, \beta, \gamma)$

$\alpha + \beta + \gamma = 0$

Now $T_{r+1} = {}^9C_r \left(2x^2\right)^{9-r} (-x)^{-r}$

$(-1)^r \, {}^9C_r 2^{9-r} x^{18-3r}$

$18 - 3r = 0 \Rightarrow r = 6$

$\therefore$ coefficient of $x^0 = {}^9C_6 2^3 = \frac{9.8.7}{6}.8$

31. Intersection of line with both the planes are the same

$\Rightarrow \frac{3}{3\beta^2 + 6(1-2\alpha) + 3} = \frac{-6}{6\alpha^2 + 6(1-2\beta) + 6}$

$\Rightarrow 2(\beta - 1)^2 + 3(\alpha - 2)^2 = 0 \Rightarrow \alpha = 2, \beta = 1$.

32. Observe that the lines L_1, L_2 & L_3 are parallel to the vector $\hat{i} - \hat{j} - \hat{k}$. Also, $\Delta = 0 = \Delta_1$ & $b_1c_2 \neq b_1c_1$

Hence the three planes intersect in a line.

$P_1 = 2x + y + z + 4 = 0$

$P_2 = 0x + y - z + 4 = 0$

$P_3 = 3x + 2y + z + 8 = 0$

P_2 and P_3 gives line L_1

Vector parallel to line $L_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix}$

$= 3\hat{i} - 3\hat{j} - 3\hat{k}$

$= 3\left[\hat{i} - \hat{j} - \hat{k}\right]$

Similarly

Vector parallel to L_2, P_3 and $P_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 3 & 2 & 1 \end{vmatrix}$

$= -\hat{i} + \hat{j} + \hat{k} = -\left[\hat{i} - \hat{j} - \hat{k}\right]$

Similarly

Vector parallel to L_3, P_1 and $P_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix}$

$= (-2)\hat{i} - (-2)\hat{j} + 2\hat{k}$

$= -2\hat{i} + 2\hat{j} + 2\hat{k}$

We can see all the lines are parallel to vector

$\left(\hat{i} - \hat{j} - \hat{k}\right)$ Also $2x + y + z = -4$

$0x + y - z = -4$

$3x + 2y + z = -8$

$\Delta = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix}$

$\Rightarrow 2(1+2) - 1(0+3) + 1(0-3)$

$\Rightarrow 2(3) - 3 - 3 = 0$ $\Delta_1 = \begin{vmatrix} -4 & 1 & 1 \\ -4 & 1 & -1 \\ -8 & 2 & 1 \end{vmatrix} = 0$

$\Delta_2 = \begin{vmatrix} 2 & -4 & 1 \\ 0 & -4 & -1 \\ 3 & -8 & 1 \end{vmatrix} = 0$

$$= -4\begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix} = 0$$

$\Delta_3 = 0$

So all planes intersection in line L.

33. Intersection of line when both the planes are same

$$\frac{3}{3\beta^2 + 6(1-2\alpha)+3} = \frac{-6}{6\alpha^2 + 6(1-2\beta)+6}$$

$\Rightarrow 2(\beta-1)^2 + 3(\alpha-2)^2 = 0 \Rightarrow \alpha = 2, \beta = 1$

$\therefore (\alpha, \beta, 1)$ lies on the planes

$2x + y + z = 4 (A)$

$x + y - 1 = 2(B)$ and $x - 2y = 0 (C)$

34. $\left(\left(\vec{\ell} \times \left(\vec{m} \times \vec{\ell}\right)\right) \times \left(\vec{n} \times \vec{m}\right)\right).\vec{n}$

$\left(\left(\vec{\ell}.\vec{\ell}\right)\vec{m} \times \left(\vec{n} \times \vec{m}\right)\right).\vec{n}$

$\left|\vec{\ell}\right|^2 \left(\left|\vec{m}\right|^2 \vec{n}\right).\vec{n} = \left|\vec{\ell}\right|^2 \left|\vec{m}\right|^2 \left|\vec{n}\right|^2 = 36$

$9 \times \frac{1}{4} \times \left|\vec{n}\right|^2 = 36$

$\left|\vec{n}\right| = 4$

35. Let O, $A(\vec{a}), B(\vec{b}), C(\vec{c})$ be the vertices of given tetrahedron

$V = \frac{1}{6}\left[\vec{a}\vec{b}\vec{c}\right]$

The centroid are $\frac{\vec{a}+\vec{b}}{3}, \frac{\vec{b}+\vec{c}}{3}, \frac{\vec{c}+\vec{a}}{3}, \frac{\vec{a}+\vec{b}+\vec{c}}{3}$

$V' = \frac{1}{6}\left[\frac{\vec{c}-\vec{a}}{3} \frac{\vec{c}-\vec{b}}{3} \frac{\vec{c}}{3}\right] = \frac{1}{6 \times 27}\left[\vec{a}\vec{b}\vec{c}\right] = \frac{1}{27}V$

$\therefore \quad k = 27$

36. $\vec{a}_1 = (a, b), \vec{a}_2 = \left(a - |2, b - \sqrt{5}\right), \vec{a}_3 = \left(a + 2, b - \sqrt{5}\right)$

Area of Δ, whose vertices are $\vec{a}_1, \vec{a}_2$ and $\vec{a}_3$, is

$$\Delta=\frac{1}{2}\begin{vmatrix}1 & a & b\\ 1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix}$$

$$\Rightarrow \Delta^2=\frac{1}{4}\begin{vmatrix}1 & a & b\\ 1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix}\begin{vmatrix}1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix} \text{ and}$$

$$\left|\vec{a}_1-\vec{a}_2\right|=3,\left|\vec{a}_1-\vec{a}_3\right|=3,\left|\vec{a}_2-\vec{a}_3\right|=4$$

$$\Rightarrow \Delta=\sqrt{5\times 2\times 2\times 1}=2\sqrt{5}$$

$$\left(2\sqrt{5}\right)^2=\frac{1}{4}\begin{vmatrix}1+\vec{a}_1\cdot\vec{a}_1 & 1+\vec{a}_1\cdot\vec{a}_2 & 1+\vec{a}_1\cdot\vec{a}_3\\ 1+\vec{a}_2\cdot\vec{a}_1 & 1+\vec{a}_2\cdot\vec{a}_2 & 1+\vec{a}_2\cdot\vec{a}_3\\ 1+\vec{a}_3\cdot\vec{a}_1 & 1+\vec{a}_3\cdot\vec{a}_2 & 1+\vec{a}_3\cdot\vec{a}_3\end{vmatrix}$$

$\therefore$ Required answer $= 4\times 5\times 4=80$

37. $\vec{c}\times\vec{a}=\vec{b}\Rightarrow\left|\vec{c}\times\vec{a}\right|=\left|\vec{b}\right|$

$$\Rightarrow\left|\vec{c}\right|\left|\vec{a}\right|\sin\theta=3\Rightarrow\left|\vec{c}\right|=\frac{3}{2\sin\theta}$$

$$\left|\vec{c}-\vec{a}\right|^2=\left|\vec{c}\right|^2+\left|\vec{a}\right|^2-2\vec{c}.\vec{a}$$

$$=\left|\vec{c}\right|^2+4-2\left|\vec{c}\right|\left|\vec{a}\right|\cos\theta$$

$$=\frac{9}{4}\operatorname{cosec}^2\theta+4-6\cot\theta$$

$$=\frac{9}{4}+\left(\frac{3}{2}\cot\theta-2\right)^2$$

$$\Rightarrow\left|\vec{c}-\vec{a}\right|^2\ge\frac{9}{4}\Rightarrow\left|\vec{c}-\vec{a}\right|\ge\frac{3}{2}$$

$\therefore$ Least value = 1.50

38. $\vec{p}=\frac{\vec{a}}{2},\vec{Q}=\frac{\vec{b}}{2},\vec{R}=\frac{\vec{c}}{2},\vec{G}=\frac{\vec{a}+\vec{b}+\vec{c}}{4}$

equation of

$$\overline{GR}\Rightarrow r=\frac{\vec{c}}{2}+\lambda\left(\frac{\vec{a}+\vec{b}+\vec{c}}{4}-\frac{\vec{c}}{2}\right)$$

equation of $\overline{PQ}\Rightarrow\vec{r}=\frac{\vec{a}}{2}+\mu\left(\frac{\vec{b}}{2}-\frac{\vec{a}}{2}\right)$

$$\text{S.D.} = \left|\frac{\left(\frac{\vec{a}}{2}-\frac{\vec{c}}{2}\right).\left(\vec{p}\times\vec{q}\right)}{\left|\vec{p}\times\vec{q}\right|}\right| = \frac{1}{2\sqrt{6}};$$

Where $\vec{a}=\hat{i},\vec{b}=\hat{j},\vec{c}=\hat{k}$

$$\vec{p}=\left(\frac{\vec{a}+\vec{b}+\vec{c}}{4}-\frac{\vec{c}}{2}\right),\ \vec{q}=\left(\frac{\vec{b}}{2}-\frac{\vec{a}}{2}\right)$$

39. $\vec{p}.\vec{q}=0,\vec{s}\times\vec{q}=\vec{p}-\vec{s}$

Let $\vec{s}=x\vec{p}+y\vec{q}+z\left(\vec{p}\times\vec{q}\right)$

$$\left(x\vec{p}+y\vec{q}+z\left(\vec{p}\times\vec{q}\right)\right)\times\vec{q}=\vec{p}-x\vec{p}-y\vec{q}-z\left(\vec{p}\times\vec{q}\right)$$

$$x\left(\vec{p}\times\vec{q}\right)+z\left[\left(\vec{p}.\vec{q}\right)\vec{q}-\left(\vec{q}.\vec{q}\right)\vec{p}\right]=(1-x)\vec{p}$$

$$-y\vec{q}-z\left(\vec{p}\times\vec{q}\right)$$

$$x\left(\vec{p}\times\vec{q}\right)-\vec{p}z=(1-x)\vec{p}-\vec{q}y-z\left(\vec{p}\times\vec{q}\right)$$

Comparing x = $x=-z, y=0, z=x-1$

$$\Rightarrow x=\frac{1}{2}, y=0, z=-\frac{1}{2}$$

$$\Rightarrow \vec{s}=\frac{\vec{p}-\vec{p}\times\vec{q}}{2}$$

$$\Rightarrow \left|\vec{s}+\vec{p}\times\vec{q}\right|=\left|\frac{\vec{p}-\vec{p}\times\vec{q}}{2}+\vec{p}\times\vec{q}\right|$$

$$=\frac{\left|\vec{p}+\vec{p}\times\vec{q}\right|}{2}=\frac{\sqrt{\left|\vec{p}\right|^2+\left|\vec{p}\times\vec{q}\right|^2+0}}{2}$$

$$=\frac{1}{\sqrt{2}}=0.707$$

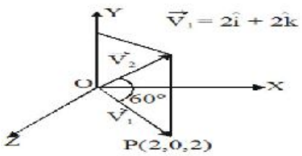
Alternat

Let $\vec{p}=\hat{i},\vec{q}=\hat{j}$ and $\vec{s}=x_1\hat{i}+h_2\hat{j}+x_3\hat{k}$

Now procced

40. Equation of plane perpendicular to XOZ plane

Is $P: x-z=0\vec{V}_2$ is coplanar with $\vec{V}_1$ and j



$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$

$\Rightarrow \vec{V}_2 = x\left(2\hat{i} + 2\hat{k}\right) + y\hat{j}$

$\because \left|\vec{V}_2\right| = \left|\vec{V}_1\right| = 2\sqrt{2}$

$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad ...(1)$

Now, $\vec{V}_2.\vec{V}_1 = \left|\vec{V}_2\right|\left|\vec{V}_1\right|\cos\frac{\pi}{3} = 4$

$4 = \left(x\left(2\hat{i} + 2\hat{k}\right) + y\hat{j}\right).\left(2\hat{i} + 2\hat{k}\right)$

$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$

$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k}\left(\because \vec{V}_2.\hat{j} > 0\right)$

Now, $\vec{V}_3 = \vec{V}_1 + x\hat{i}$

$\because \vec{V}_3.\vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$

$\Rightarrow \vec{V}_3 = 2\hat{i} + 2\hat{k} - 4\hat{i}$

$\vec{V}_3 = -2\hat{i} + 2\hat{k}$

$\Rightarrow$ volume of tetrahedron

$$V = \left|\frac{1}{6}\begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix}\right|$$

$\Rightarrow V = \frac{1}{6}8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$

41. $\left|\vec{A}\right| = 3, \left|\vec{B}\right| = 4, \left|\vec{C}\right| = 5$

$\vec{A}.\left(\vec{B} + \vec{C}\right) = \vec{B}.\left(\vec{C} + \vec{A}\right) = \vec{C}.\left(\vec{A} + \vec{B}\right) = 0$

Now, $\left|\vec{A} + \vec{B} + \vec{C}\right|^2$

$= A^2 + B^2 + C^2 + \vec{A}\left(\vec{B} + \vec{C}\right) + \vec{B}\left(\vec{C} + \vec{A}\right) + \vec{C}\left(\vec{A} + \vec{B}\right)$

$= 50 + 0 + 0 + 0 = 50 \Rightarrow \left|\vec{A} + \vec{B} + \vec{C}\right| = 5\sqrt{2}$

42. The distance of the point 'a' from the plane $\vec{r}\cdot\vec{n} = q$

Measured in the direction of the unit vector $\hat{b}$ is

$= \dfrac{q - \vec{a}\cdot\vec{n}}{\vec{b}.\vec{n}}.$

Here $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{n} = \hat{i} + \hat{j} + \hat{k}$ and $q = 5$

Also

$$\vec{b} = \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{\sqrt{(2)^2 + (3)^2 + (-6)^2}} = \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{7}$$

$\therefore$ The required distance =

$$\frac{5 - \left(\hat{i} + 2\hat{j} + 3\hat{k}\right).\left(\hat{i} + \hat{j} + \hat{k}\right)}{\frac{1}{7}\left(2\hat{i} + 3\hat{j} - 6\hat{k}\right).\left(\hat{i} + \hat{j} + \hat{k}\right)} = 7.$$

43. $$\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix} = \begin{vmatrix} 9 & 6 & -1 \\ 6 & 16 & 6 \\ -1 & 6 & 36 \end{vmatrix}$$

$= 9(576 - 36) - 6(216 + 6)$

$-(36 + 16)$

$= 4860 - 1332 - 52 = 3476$

$\therefore$ Volume of tetrahedron $= V = \dfrac{1}{6}\left[\vec{a}\vec{b}\vec{c}\right]$

$\therefore\ V^2 = \dfrac{1}{36}.\left[\vec{a}\vec{b}\vec{c}\right]^2 = \dfrac{1}{36} \times 3476$

$= \dfrac{869}{9} = \dfrac{11 \times 79}{3^2} = \dfrac{pq}{r^2}$

$\therefore\quad \dfrac{(p+q)}{r} = \dfrac{11 + 79}{3} = 30$

44. $\bar{b} = \dfrac{-\bar{a}}{2} + \dfrac{\sqrt{3}}{2}\hat{k};\quad \bar{b} = \dfrac{-1}{2\sqrt{2}}\hat{i} - \dfrac{1}{2\sqrt{2}}\hat{j} + \dfrac{\sqrt{3}}{2}\hat{k}$

If $\left|\bar{b} \times \bar{c}\right| = \left|\bar{a} \times \bar{c}\right| \Rightarrow$ then $\bar{c}$ is perpendicular to both $\bar{a}$ and $\bar{b}$

$\bar{c} = \lambda\left(\bar{a} \times \bar{b}\right) \,\&\, \left|\bar{c}\right| = 1$

$\therefore\ \bar{c} = \pm\dfrac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}$

45. $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = \dfrac{1}{2}$

$\left|\vec{a}\right| = \left|\vec{b}\right| = \left|\vec{c}\right| = 1$

$\left[\vec{a}\,\vec{b}\,\vec{c}\right] = p + \dfrac{q}{2} + \dfrac{r}{2} \ldots.(1)$

$\left[\vec{a}\,\vec{b}\,\vec{c}\right] = \dfrac{p}{2} + \dfrac{q}{2} + r \ldots.(2)$

$0 = \dfrac{p}{2} + q + \dfrac{r}{2} \ldots..(3)$

$(1)-(2) \Rightarrow p = r$

& from (3) q = -p

$\therefore \text{from}\,(2)\,\dfrac{p}{2} + \dfrac{q}{2} + r = \left[\vec{a}\,\vec{b}\,\vec{c}\right] \Rightarrow r = \left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$\therefore p^2 + q^2 + r^2 = 3\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2$

$$= 3\begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix} = 3\begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= 3\left[1 + 2.\frac{1}{2}.\frac{1}{2}.\frac{1}{2} - \frac{1^2}{2^2} - \frac{1^2}{2^2} - \frac{1}{2^2}\right]$$

$$= = 3\left[1 + \frac{1}{4} - \frac{3}{4}\right] = 3\left[1 - \frac{1}{2}\right] = \frac{3}{2}$$

46. $\left|\vec{a} + \vec{b} + \vec{c}\right|^2 \geq 0$

$\therefore \vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a} \geq \dfrac{-1}{2}\left(\left|\vec{a}\right|^2 + \left|\vec{b}\right|^2 + \left|\vec{c}\right|^2\right)$

Given. $\left|\vec{a} - \vec{b}\right|^2 + \left|\vec{b} - \vec{c}\right|^2 + \left|\vec{c} - \vec{a}\right|^2 = 9$

$\Rightarrow \left|\vec{a}\right|^2 + \left|\vec{b}\right|^2 - 2\vec{a}.\vec{b} + \left|\vec{b}\right|^2 + \left|\vec{c}\right|^2 - 2\vec{b}.\vec{c} + \left|\vec{c}\right|^2 + \left|\vec{a}\right|^2$

$-2\vec{c}\,\vec{a} = 9$

$\Rightarrow \vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}=-\frac{3}{2}$ ....(i)

Also,

$\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}\geq\frac{-1}{2}\left(\left|\vec{a}\right|^2+\left|\vec{b}\right|^2+\left|\vec{c}\right|^2\right)\geq-\frac{3}{2}$....(ii)

From Eqs. (i) and (ii), $\left|\vec{a}+\vec{b}+\vec{c}\right|=0$

$\Rightarrow \vec{a}+\vec{b}+\vec{c}=\vec{0}$

$\left|2\vec{a}+5\vec{b}+5\vec{c}\right|=\left|2\vec{a}-5\vec{a}\right|=3$

47. $\vec{c}.\vec{a}=x\vec{a}.\vec{a}+y\vec{b}.\vec{a}+\left(\vec{a}\times\vec{b}\right).\vec{a}$

$\Rightarrow x=2\cos\alpha$

Similarly, $y=2\cos\alpha$

$\left|\vec{c}\right|^2=\left(4\cos^2\alpha\right)\times4+\left(4\cos^2\alpha\right)\times4+16$

$\Rightarrow \cos^2\alpha=0$

48. $A(1,0,0), B\left(\frac{1}{2},\frac{1}{2},0\right) \& C\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$

$\overrightarrow{OA}=\hat{i}, \overrightarrow{OB}=\frac{1}{2}\hat{i}+\frac{1}{2}\hat{j},$

$\overrightarrow{OC}=\frac{1}{3}\hat{i}+\frac{1}{3}\hat{j}+\frac{1}{3}\hat{k}$

$\Rightarrow V=\frac{1}{36}$

49. Orthocentre is foot of perpendicular drawn from origin on the plane.

50. $x+y+z<8, x,y,z\geq1 \Rightarrow x+y+z+w=8$

$w\geq1$

$\Rightarrow$ No of Solutions$=n={}^7C_3=35$

51. The direction ratios RS are3-1,5-2,7-3i.e.2,3,4 so direction cosines of RS $\frac{2}{\sqrt{29}},\frac{3}{\sqrt{29}},\frac{4}{\sqrt{29}}$ and the projection of PQ on is RS

52. Equation of plane passing through the intersection of the planes $2x-5y+z=3$ and $x+y+4z=5$ is $(2x-5y+z-3)+\lambda(x+y+4z-5)=0$

$(2+\lambda)x+(-5+\lambda)y+(1+4\lambda)-3-5\lambda=0...(i)$.

Which is parallel to the plane $x+3y+6z=1$

Then $\dfrac{2+\lambda}{1}=\dfrac{-5+\lambda}{3}=\dfrac{1+4\lambda}{6}$

$\therefore = \dfrac{-11}{2}$

From eq. (i)

$-\dfrac{7}{2}\times-\dfrac{21}{2}y-21z+\dfrac{49}{2}=0$

$\therefore x+3y+6z=7$

Hence $k=7$

53. Clearly, a = 1, b = 2, c = 3.

The second line can be written as

$\dfrac{x-a}{a}=\dfrac{y}{0}=\dfrac{z}{c}$ .....(1)

Now equation of plane through the line

$\dfrac{y}{b}+\dfrac{z}{c}=1$, $x=0$ is $\left(\dfrac{y}{b}+\dfrac{z}{c}-1\right)+\lambda x=0$

Or $\lambda x+\dfrac{y}{b}+\dfrac{z}{c}=1$ ....(2)

$\because$ plane (2) is parallel to the line (1)

$\Rightarrow \lambda.a+\dfrac{1}{6}.0+\dfrac{1}{c}.c=0 \;\Rightarrow\; \lambda=-\dfrac{1}{a}$

$\therefore$ equation of plane Π is

$-\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$ or $-\dfrac{x}{1}+\dfrac{y}{2}+\dfrac{z}{3}=1$

$\therefore$ $A(-1,0,0), B(0,2,0), C(0,0,3)$

$\therefore$ volume of tetrahedron

$OABC=\left|\dfrac{1}{6}\left[-\hat{i}\,2\hat{j}\,3\hat{k}\right]\right|=6.\dfrac{1}{6}\left[\hat{i}\hat{j}\hat{k}\right]=1$

54. Let O, $A(\vec{a}), B(\vec{b}), C(\vec{c})$ be the vertices of given tetrahedron

$V=\dfrac{1}{6}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

The centroid are $\dfrac{\vec{a}+\vec{b}}{3},\dfrac{\vec{b}+\vec{c}}{3},\dfrac{\vec{c}+\vec{a}}{3},\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

$$V' = \frac{1}{6}\left[\frac{\vec{c}-\vec{a}}{3}\,\frac{\vec{c}-\vec{b}}{3}\,\frac{\vec{c}}{3}\right] = \frac{1}{6\times 27}\left[\vec{a}\vec{b}\vec{c}\right] = \frac{1}{27}V$$

$\therefore k = 27$

55.

56. $D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & a \end{vmatrix} = a$

Unique solution $a \neq 0$

Ordered pair = 6×7

57. Given plane can be

$(2x + 3y - z + 1) + \lambda(x + y - 2z + 3) = 0$

$\Rightarrow x(2+\lambda) + y(3+\lambda) - z(1+2\lambda) + 1 + 3\lambda = 0$

It is perpendicular to $3x - y - 2z - 4 = 0$

$\Rightarrow 3(2+\lambda) - (3+\lambda) + 2(1+2\lambda) = 0$

$\Rightarrow 6\lambda + 5 = 0 \Rightarrow \lambda = -\frac{5}{6}$

$\therefore$ Plane is $7x + 13y + 4z - 9 = 0$

$\therefore\ \ell = 7 \quad m = 4 \quad n = -9$

$\frac{\ell - n}{m} = \frac{16}{4} = 4.00$

58. $(15\lambda + 1, 16\lambda - 1, n\lambda + 1)$ point

Then $n\lambda + 1 = 0$

$6(15\lambda + 1)^2 + 5(16\lambda - 1)^2 = 1$

$\Rightarrow 6\left(1 - \frac{15}{n}\right)^2 + 5\left(-1 - \frac{16}{n}\right)^2 = 20$

$\Rightarrow 6(n - 15)^2 + 5(n + 16)^2 = 20n^2$

$\Rightarrow -9n^2 - 20n + 2630 = 0$

59. $\frac{x-3}{3} = \frac{y-1}{4} = \frac{z+1}{0} = \lambda$

$\bullet A(0, 1, -1)$

$\bullet B(3,5,-1)$

General Point on line is $(3\lambda, 4\lambda+1, -1)$

This point will satisfy plane $x+y-2z=5$

$3\lambda+4\lambda+1+2=5$

$\lambda=\frac{2}{7}$

$P\left(\frac{6}{7},\frac{15}{7},-1\right)$

60. $6x+3y+2z-6=0$

$P(x_1,y_1,z_1)$

$A(0,y_1,z_1), B(x_1,0,z_1), C(x_1,y_1,0)$

$$\vec{N}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ x_1 & -y_1 & 0\\ 0 & y_1 & -z_1\end{vmatrix}$$

$=(y_1z_1)\hat{i}+(x_1z_1)\hat{j}+(x_1y_1)\hat{k}$

$\Rightarrow$ Equation of plane containing points A,

B & C

$\frac{x}{x_1}+\frac{y}{y_1}+\frac{z}{z_1}=2 \qquad \ldots(1)$

Let $P(\alpha,\beta,\gamma)$ be the foot of perpendicular from (0, 0, 0) to the plane (1)

$\Rightarrow \frac{\alpha}{1/x_1}=\frac{\beta}{1/y_2}=\frac{\gamma}{1/z_1}=\lambda$

$x_1=\frac{\lambda}{\alpha}, y_1=\frac{\lambda}{\beta}, z_1=\frac{\lambda}{\gamma}$

(x_1,y_1,z_1) lies in $6x+3y+2z-6=0$

$\Rightarrow \frac{6\lambda}{\alpha}+\frac{3\lambda}{\beta}+\frac{2\lambda}{\gamma}=6 \qquad \ldots(2)$

(α,β,γ) lies in (1)

$\Rightarrow \frac{\alpha^2}{\lambda}+\frac{\beta^2}{\lambda}+\frac{\gamma^2}{\lambda}=2 \qquad \ldots(3)$

From (2) & (3)

$$\left(\alpha^2+\beta^2+\gamma^2\right)\left(\frac{6}{\alpha}+\frac{3}{\beta}+\frac{2}{\gamma}\right)=12$$

$$\Rightarrow \text{locus is } \left(x^2+y^2+z^2\right)\left(\frac{1}{x}+\frac{1}{2y}+\frac{1}{3z}\right)=2$$

61. $x+y-z=1$

$(m-1)x+(3m-7)y-5z=-3$

$(m-2)x+(2m-5)y=-4$

$\because$ all the three planes intersects at a line

$$\because \begin{vmatrix} 1 & 1 & -1 \\ m-1 & 3m-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix}=0$$

$\Rightarrow m=3,-2$

at $m=3$ $x+y-z=1$

$2x+2y-5z=-3$

$x+y=-4$

System does not have any solution but at m = -2

$P_1: x+y-z=1$

$P_2: 3x+13y+5z=3$

$P_3: 4x+9y=4$

$\Rightarrow 5P_1+P_2=2P_3$

$\Rightarrow$ direction ratio of line of intersection are given by

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 13 & 5 \end{vmatrix}=18\hat{i}-8\hat{j}+10\hat{k}$$

$\equiv \langle 9,-4,5 \rangle$

$\Rightarrow \alpha=9\lambda, b=-4\lambda, c=5\lambda$

62. Any point on the line of intersection is $(\lambda+2,-1,-\lambda)$ For maximum distance

$$=\left((\lambda+2-3)\hat{i}+(-1+4)\hat{j}+(-\lambda-5)\hat{k}\right).\left(\hat{i}-\hat{k}\right)=0$$

$\Rightarrow \lambda=-2$

Normal $\vec{n}=3\hat{i}+5\hat{j}+3\hat{k}$

equation of plane will be $3x+5y+3z=1$

63. $L_1 : \frac{x+2}{2} = \frac{y+6}{3} = \frac{z-34}{-10}$

and $L_2 : \frac{x+6}{4} = \frac{y-7}{-3} = \frac{z-7}{-2}$

are skew lines as $\begin{vmatrix} -2+6 & -6-7 & 34-7 \\ 2 & 3 & -10 \\ 4 & -3 & -2 \end{vmatrix} \neq 0$

Direction of line perpendicular to $L_1 \,\&\, L_2$

is $\begin{vmatrix} i & j & k \\ 2 & 3 & -10 \\ 4 & -3 & -2 \end{vmatrix} = (-36i - 36j - 18k)$

so || vector is $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$

Let line perpendicular to $L_1 \,\&\, L_2$ is L

with A and B points respectively on $L_1 \,\&\, L_2$

$A(2\lambda - 2, 3\lambda - 6, -10\lambda + 34)$

$B(4\mu - 6, -3\mu + 7, -2\mu + 7)$

AB is parallel to $2\hat{i} + 2\hat{j} + \hat{k}$

$\therefore \frac{2\lambda - 4\mu + 4}{2} = \frac{3\lambda + 3\mu - 13}{2} = \frac{2\mu - 10\lambda + 27}{1}$

Solving these,

$2\lambda - 4\mu + 4 = 3\lambda + 3\mu - 13$

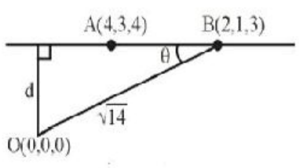
$\Rightarrow \lambda + 7\mu = 17 \;\; \ldots .(1)$

and $3\lambda + 3\mu - 13 = 4\mu - 20\lambda + 54$

$\Rightarrow 23\lambda - \mu = 67 \;\; \ldots ..(2)$

solving (1) and (2), $\lambda = 3$ and $\mu = 2$

$\therefore A \equiv (4,3,4)$ and $B(2,1,3)$

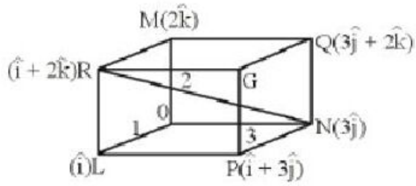


$\cos\theta = \frac{\overrightarrow{AB}.\overrightarrow{OB}}{|\overrightarrow{AB}||\overrightarrow{OB}|}$

$$=\frac{4+2+3}{3.\sqrt{14}}=\frac{3}{\sqrt{14}}$$

$$d=OB\sin\theta=\sqrt{14}.\frac{\sqrt{5}}{\sqrt{14}}\Rightarrow d=\sqrt{5}\Rightarrow d^2=5$$

64.



equation of OL : $\vec{r}=0+\mu\hat{i}$

equation of RN : $\vec{r}=3\hat{j}+\lambda\left(\hat{i}-3\hat{j}+2\hat{k}\right)$

(skew body diagonal)

$\Rightarrow \ell=S.D=\left|\left(3\hat{j}\right).\hat{n}\right|,\vec{n}=\left(\hat{i}-3\hat{j}+2\hat{k}\right)\times\hat{i}$

$$=\left|3\hat{j}.\frac{\left(2\hat{j}+3\hat{k}\right)}{\sqrt{13}}\right|=\frac{6}{\sqrt{13}}$$

equation of ON : $\vec{r}=0+\mu\left(3\hat{j}\right)$

equation of QL : $\vec{r}=\hat{i}+\lambda\left(\hat{i}-3\hat{j}-2\hat{k}\right)$

(skew body diagonal)

$\Rightarrow n=S.D=\left|\hat{i}.\hat{n}\right|,\vec{n}=\left(\hat{i}-3\hat{j}-2\hat{k}\right)\times\hat{j}$

$$=\left|\frac{\hat{i}.\left(-2\hat{i}-\hat{k}\right)}{\sqrt{5}}\right|=\frac{2}{\sqrt{5}}$$

equation of OM : $\vec{r}=0+\lambda\left(2\hat{k}\right)$

equation RN : $\vec{\gamma}=3\hat{j}+\lambda\left(\hat{i}-3\hat{j}+2\hat{k}\right)$

(skew body diagonal)

m = S.D. = $\left|3\hat{j}.\hat{n}\right|,\vec{n}=\left(\hat{i}-3\hat{j}+2\hat{k}\right)\times\hat{k}$

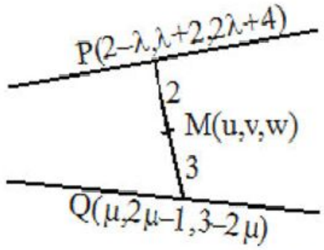
$$=\left|3\hat{j}\frac{\left(-3\hat{i}-\hat{j}\right)}{\sqrt{10}}\right|=\frac{3}{\sqrt{10}}$$

$\Rightarrow \sqrt{13\ell^2 + 10m^2 + 5n^2} = 7$

65.

66. $L_1 : \frac{x-2}{-1} = \frac{y-2}{1} = \frac{z-4}{2}$ and

$L_2 : \frac{x-0}{1} = \frac{y+1}{2} = \frac{z-3}{-2}$



Using section formula and eliminating

λ, μ we get $w + 2u = 6$

M(u, v, w) moves on

$2x + z = 6$

67. Using V.T.P

we get $\vec{r}.\vec{a} = \vec{r}.\vec{b} = \vec{r}.\vec{c}$

$\Rightarrow \vec{r} = \pm\frac{1}{\sqrt{3}}\left(\vec{a} + \vec{b} + \vec{c}\right)$

68. $\vec{r} = 12\vec{a} + 5\vec{b} + 13\vec{c}$

$\lambda = \vec{r}.\left(\frac{\vec{a}+\vec{b}+\vec{c}}{\sqrt{3}}\right) = \frac{30}{\sqrt{3}}$

69-71 $L_1 : \frac{x}{0} - \frac{y-b}{-b} = \frac{z}{c}$

$L_2 : \frac{x-a}{a} - \frac{y}{0} = \frac{z}{c}$

Equation of plane p: $\begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix} = 0$

$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$

Lines L_1 and L_2 in vector form

$L_1 = \vec{r} = \overrightarrow{p_1} + \lambda\overrightarrow{q_1} \Rightarrow \vec{r} = b\hat{j} + \lambda\left(-b\hat{j} + c\hat{k}\right)$ and $L_2 = \vec{r} = \overrightarrow{p_2} + \lambda\overrightarrow{q_2} \Rightarrow \vec{r} = a\hat{i} + 4\left(a\hat{i} + c\hat{k}\right)$

Shortest distance $= \dfrac{\left[\overrightarrow{p_1} - \overrightarrow{p_2}\ \overrightarrow{q_1}\ \overrightarrow{q_2}\right]}{\left|\overrightarrow{q_1}\times\overrightarrow{q_2}\right|} = \dfrac{\begin{vmatrix} -a & b & 0 \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}}{\sqrt{(bc)^2 + (ac)^2 + (ab)^2}} = \dfrac{1}{4}$

$\Rightarrow \left|\dfrac{2abc}{\sqrt{(bc)^2 + (ca)^2 + (ab)^2}}\right| = \dfrac{1}{4} \Rightarrow \dfrac{1}{a^2} + \dfrac{1}{b^2} + \dfrac{1}{c^2} = 64$

72. T is an equilateral triangle with the vertices at $(1,0,0),(0,1,0)\, and\, (0,0,1)$

$\Rightarrow$ area of the region T is $\dfrac{\sqrt{3}}{2}$

73. Take a point $P\left(\dfrac{1}{2},\dfrac{1}{3},\dfrac{1}{6}\right)$ on the plane $x + y + z = 1$. The region S is shown as shaded region. Area of S= $\dfrac{\sqrt{3}}{2} - \dfrac{\sqrt{3}}{4}\left(a^2 + b^2 + c^2\right)$, where a, b , c are sides of three small equilateral triangles

$\dfrac{7\sqrt{3}}{36}$

74. $\Rightarrow (1+\lambda)x + (2-\lambda)y + (1+2\lambda)z - 1 + \lambda = 0$

$\perp$ to $x + y + z = 5$

$1 + \lambda + 2 - \lambda + 1 + 2\lambda = 0 \Rightarrow \lambda = -2$

$\therefore P: -x + 4y - 3z - 3 = 0$

$x - 4y + 3z = 0$

75. $Q(1,1,2)$

$\therefore 1 + \lambda + 2 - \lambda + 2 + 4\lambda - 1 + \lambda = 0$

$\lambda = -\dfrac{4}{5}$

$\therefore P = \dfrac{1}{5}x + \dfrac{14y}{5} - \dfrac{3z}{5} - \dfrac{9}{4} = 0$

$x + 14y - 3z = 9$

$\vec{r} = \lambda\left(\hat{i} + 14\hat{j} - 3\hat{k}\right)$

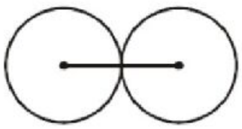
76. B_1, B_2 both touches xz plane

$\Rightarrow$ Centre of B_3 lies on x- axis

Let centre be $(\alpha, 0, 0)$ having radius 5

$$\left(\alpha - \frac{5}{\sqrt{3}}\right) + 25 + 0 = 100 \Rightarrow \alpha = \frac{20}{\sqrt{3}}, \beta = C$$

$$\Rightarrow \alpha + \beta = \frac{20}{\sqrt{3}}$$

77. $\left(\frac{5}{\sqrt{3}}, 5, 0\right)$  $\left(\frac{20}{\sqrt{3}}, 0, 0\right)$

$$B_1\left(\frac{25}{2\sqrt{3}}, \frac{5}{2}, 0\right)B_3$$

Plane passing through $\left(\frac{25}{2\sqrt{3}}, \frac{5}{2}, 0\right)$ having normal vector $\left(\frac{15}{\sqrt{3}}\hat{i} - 5\hat{j}, +0\hat{k}\right)$ hence the equation of plane

$$\left(x - \frac{25}{2\sqrt{3}}\right)\frac{15}{\sqrt{3}} + \left(y - \frac{5}{2}\right)(-5) + (z - 0)0 = \frac{15x}{\sqrt{3}} - \frac{125}{2} - 5y + \frac{25}{2} = 0$$

$$\Rightarrow \frac{15x}{\sqrt{3}} - 5y = 50 \Rightarrow \sqrt{3x} - y = 10$$

78. Given $M(1, 2, 3)$

$N(2 + \lambda, 1 - 2\lambda, \lambda - 1)$

$\therefore \overrightarrow{MN} = p.v. of\ N - p of M$

$= (\lambda + 1)\hat{i} - (1 + 2\lambda)\hat{j} + (\lambda - 4)\hat{k}$

Also $\vec{n}$ = normal vector of plane $= i - 4j + 3k$

As $\overrightarrow{MN} \cdot \vec{n} = 0$

$\Rightarrow 1(\lambda + 1) + 4(1 + 2\lambda) + 3(\lambda - 4) = 0$

$$\Rightarrow \lambda = \frac{7}{12}$$

79. The normal vector of plane passing through $P(5, -5, 2)$ and $Q\left(\frac{10}{3}, \frac{-10}{3}, \frac{4}{3}\right)$ and perpendicular to plane

$\vec{r}.\left(\hat{i}-\hat{j}+\hat{k}\right)+1=0$

$\vec{n}=\overrightarrow{MN}\times\left(\hat{i}-\hat{j}+\hat{k}\right)$

$=\frac{1}{3}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -5 & 5 & -2\\ 1 & -1 & 1\end{vmatrix}=\hat{i}+\hat{j}$

$\therefore$ The equation of required plane is $1(x-5)+1$

$(y+5)+0(z-2)=0$

$x+y=0\,or\,\vec{r}.\left(\hat{i}+\hat{j}\right)=0$

80. (P) $\vec{r}$ cannot be null vector.

Also, $\vec{r}\times\hat{i}$ is perpendicular to $\hat{i}$.

Hence $\vec{r}\times\hat{i}\neq\hat{i}$

(Q) $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1\Rightarrow x+y+z=1$ (A plane)

so, length of perpendicular distance of

origin from this plane = $|\vec{r}|=\frac{1}{\sqrt{3}}$

Hence, there is only one point i.e. foot of normal drawn from the origin to the plane, satisfying the given equation.

(R) There are infinite points in the space which lies at unit distance from (1, 0, 0)

81. $(P)=\bar{a}=\left(3\hat{i}-4\hat{j}+2\hat{k}\right)$

$\bar{b}=-4\hat{i}-7\hat{j}+4\hat{k}$

Projection of $\bar{a}$ on $\bar{b}$

$=\frac{\bar{a}.\bar{b}}{|\bar{b}|}=\frac{-12+28+8}{\sqrt{81}}=\frac{24}{\sqrt{81}}=\frac{8}{3}$

(Q) $\bar{\alpha}+\bar{\beta}+\bar{\gamma}=0$

$\bar{\alpha}+\bar{\beta}=-\bar{\gamma}$

$\Rightarrow\left(\bar{\alpha}\right)^2+\left(\bar{\beta}\right)^2+2\bar{\alpha}.\bar{\beta}=\left(\bar{\gamma}\right)^2$

$\Rightarrow 2.3.5\cos\theta=49-9-25=15$

$\cos\theta=\frac{1}{2}$.

(R) $\left(\overline{A}+\overline{B}\right)\times\left(\overline{A}-\overline{B}\right)=\overline{A}\times\overline{A}-\overline{A}\times\overline{B}+\overline{B}\times\overline{A}-\overline{B}\times\overline{B}$

$=-2\left(\overline{A}\times\overline{B}\right)$

$$\overline{A}\times\overline{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & -1 & -2\\ 2 & 3 & 1\end{vmatrix}=5\hat{i}-7\hat{j}+11\hat{k}$$

$\left|\overline{A}\times\overline{B}\right|=\sqrt{195}$

$\left|\left(\overline{A}\times\overline{B}\right)\times\left(\overline{A}-\overline{B}\right)\right|=2\sqrt{195}$

(S) $\left(\overline{a}+\overline{b}+\overline{c}\right)^2=\overline{a}^2+\overline{b}^2+\overline{c}^2+2\left(\overline{a}.\overline{b}+\overline{b}.\overline{c}+\overline{c}.\overline{a}\right)$

$0=3+2\left(\overline{a}.\overline{b}+\overline{b}.\overline{c}+\overline{c}.\overline{a}\right)$

$\Rightarrow\dfrac{-3}{2}$

82. (I) $\vec{r}=\lambda\vec{a}+\mu\vec{b}+\gamma\left(\vec{a}\times\vec{b}\right)$

$\vec{r}.\vec{a}=0\Rightarrow\lambda+\dfrac{\mu}{2}=0$

$\vec{r}.\vec{a}=2\Rightarrow\dfrac{\lambda}{2}+\mu=2$

$\left(\vec{r}\times\vec{a}\right).\vec{b}=1\Rightarrow\dfrac{3\gamma}{4}=1$

Solving $\lambda=\dfrac{-4}{3},\mu=\dfrac{8}{3},\gamma=\dfrac{4}{3}$

(I) (iii) (R)

(II) $\vec{r}.\vec{a}=0\Rightarrow\lambda+\dfrac{\mu}{2}=0$

$\vec{r}.\vec{b}=1\Rightarrow\dfrac{\lambda}{2}+\mu=1$

$\gamma=0;\lambda=\dfrac{-2}{3}\mu=\dfrac{4}{3}$

(II) (ii) (P)

(III) $\vec{r}\times\vec{a}=3\vec{a}\times\vec{b}$

$\left(\vec{r}+3\vec{b}\right)\times\vec{a}=0$

$\vec{r}+3\vec{b}=\lambda\vec{a}$

$\vec{r} = \lambda\vec{a} - 3\vec{b}$

$\vec{r}.\vec{a} = 1 \Rightarrow \lambda\frac{-3}{2} = 1 \Rightarrow \lambda = \frac{5}{2}$

$\vec{r} = \frac{5}{2}\vec{a} - 3\vec{b}$

$\lambda = \frac{5}{2}, \mu = -3, \lambda = 0$

(III) (iv) (P)

(IV) $\vec{a} \times (\vec{b} \times \vec{r}) = \vec{a} \times \vec{b}$

$(\vec{a}.\vec{r})\vec{b} - (\vec{a}.\vec{b})\vec{r} = \vec{a} \times \vec{b}$

$2\vec{b} - \frac{1}{2}\vec{r} = \vec{a} \times \vec{b}$

$\vec{r} = 4\vec{b} - 2(\vec{a} \times \vec{b})$

$\lambda = 0, \mu = 4, r = 2$

(IV) (iii) (S)

83. (I) $\vec{r} = \lambda\vec{a} + \mu\vec{b} + \gamma(\vec{a} \times \vec{b})$

$\vec{r}.\vec{a} = 0 \Rightarrow \lambda + \frac{\mu}{2} = 0$

$\vec{r}.\vec{a} = 2 \Rightarrow \frac{\lambda}{2} + \mu = 2$

$(\vec{r} \times \vec{a}).\vec{b} = 1 \Rightarrow \frac{3\gamma}{4} = 1$

Solving $\lambda = \frac{-4}{3}, \mu = \frac{8}{3}, \gamma = \frac{4}{3}$

(I) (iii) (R)

(II) $\vec{r}.\vec{a} = 0 \Rightarrow \lambda + \frac{\mu}{2} = 0$

$\vec{r}.\vec{b} = 1 \Rightarrow \frac{\lambda}{2} + \mu = 1$

$\gamma = 0; \lambda = \frac{-2}{3}\mu = \frac{4}{3}$

(II) (ii) (P)

(III) $\vec{r} \times \vec{a} = 3\vec{a} \times \vec{b}$

$(\vec{r} + 3\vec{b}) \times \vec{a} = 0$

$\vec{r} + 3\vec{b} = \lambda\vec{a}$

$\vec{r} = \lambda\vec{a} - 3\vec{b}$

$\vec{r}.\vec{a} = 1 \Rightarrow \lambda\frac{-3}{2} = 1 \Rightarrow \lambda = \frac{5}{2}$

$\vec{r} = \frac{5}{2}\vec{a} - 3\vec{b}$

$\lambda = \frac{5}{2}, \mu = -3, \lambda = 0$

(III) (iv) (P)

(IV) $\vec{a} \times (\vec{b} \times \vec{r}) = \vec{a} \times \vec{b}$

$(\vec{a}.\vec{r})\vec{b} - (\vec{a}.\vec{b})\vec{r} = \vec{a} \times \vec{b}$

$2\vec{b} - \frac{1}{2}\vec{r} = \vec{a} \times \vec{b}$

$\vec{r} = 4\vec{b} - 2(\vec{a} \times \vec{b})$

$\lambda = 0, \mu = 4, r = 2$

(V) (iii) (S)

84. (I) $\vec{r} = \lambda\vec{a} + \mu\vec{b} + \gamma(\vec{a} \times \vec{b})$

$\vec{r}.\vec{a} = 0 \Rightarrow \lambda + \frac{\mu}{2} = 0$

$\vec{r}.\vec{a} = 2 \Rightarrow \frac{\lambda}{2} + \mu = 2$

$(\vec{r} \times \vec{a}).\vec{b} = 1 \Rightarrow \frac{3\gamma}{4} = 1$

Solving $\lambda = \frac{-4}{3}, \mu = \frac{8}{3}, \gamma = \frac{4}{3}$

(I) (iii) (R)

(II) $\vec{r}.\vec{a} = 0 \Rightarrow \lambda + \frac{\mu}{2} = 0$

$\vec{r}.\vec{b} = 1 \Rightarrow \frac{\lambda}{2} + \mu = 1$

$\gamma = 0; \lambda = \frac{-2}{3}\mu = \frac{4}{3}$

(II) (ii) (P)

(III) $\vec{r} \times \vec{a} = 3\vec{a} \times \vec{b}$

$\left(\vec{r}+3\vec{b}\right)\times\vec{a}=0$

$\vec{r}+3\vec{b}=\lambda\vec{a}$

$\vec{r}=\lambda\vec{a}-3\vec{b}$

$\vec{r}.\vec{a}=1\Rightarrow\lambda\frac{-3}{2}=1\Rightarrow\lambda=\frac{5}{2}$

$\vec{r}=\frac{5}{2}\vec{a}-3\vec{b}$

$\lambda=\frac{5}{2},\mu=-3,\lambda=0$

(III)(iv) (P)

(IV) $\vec{a}\times\left(\vec{b}\times\vec{r}\right)=\vec{a}\times\vec{b}$

$\left(\vec{a}.\vec{r}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{r}=\vec{a}\times\vec{b}$

$2\vec{b}-\frac{1}{2}\vec{r}=\vec{a}\times\vec{b}$

$\vec{r}=4\vec{b}-2\left(\vec{a}\times\vec{b}\right)$

$\lambda=0,\mu=4,r=2$

(iv)(iii) (S)

85-86. (I) $\left[\vec{a}\vec{b}\vec{c}\right]^2=\begin{vmatrix}1&1&1\\1&-1&1\\1&2&-1\end{vmatrix}=16$

(II) $\vec{n}=\vec{r}_2\times\vec{r}_3=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&3&4\\2&1&-2\end{vmatrix}$

$=\hat{i}(-6-4)-\hat{j}(-2-8)+\hat{k}(1-6)$

$=-10\hat{i}+10\hat{j}-5\hat{k}\Rightarrow\hat{n}=\frac{2\hat{i}-2\hat{j}+\hat{k}}{3}$

$\therefore\ d=\left|\text{Projection of }\overline{OP}\text{ on }\vec{n}\right|$

$=\left|\frac{\left(3\hat{i}-2\hat{j}-\hat{k}\right).\left(2\hat{i}-2\hat{j}+\hat{k}\right)}{3}\right|$

$=\frac{6+4-1}{3}=3$ units

(III) $L_1:\frac{x-0}{1}=\frac{y+1}{1}=\frac{z-0}{1}=\lambda$

$L_2:\frac{x+1}{2}=\frac{y-0}{1}=\frac{z-0}{1}=\mu$

Hence any point on L_1 and L_2 can be

$(\lambda,\lambda-1,\lambda)$ and $(2\mu-1,\mu,\mu)$

$\therefore\ \frac{2\mu-1-\lambda}{2}=\frac{\mu-\lambda+1}{1}=\frac{\mu-\lambda}{1}$

Solving $\mu=1$ and $\lambda=3$

A = (3, 2, 3) and B = (1, 1, 1)

$\Rightarrow AB=\sqrt{4+1+4}=3$

(IV) $V\ell=\frac{1}{6}\left[\vec{a}\vec{b}\vec{c}\right];\ V_s=\frac{1}{6}.\frac{1}{27}\left[\vec{a}\vec{b}\vec{c}\right]$

Hence $\frac{V_s}{V\ell}=\frac{1}{27}=\frac{m}{n}\text{or}\frac{n}{27}=\frac{m}{1}=k$

$\therefore$ m and n are relatively prime

$\Rightarrow k=1,(m+n)=28$

87. Plane through line $y=ax,z=c\ is\left(y-ax\right)+\lambda_1\left(z-c\right)=0$

Similarly $\left(y+ax\right)+\lambda_2\left(z+c\right)=0$

$\Rightarrow\left(y+c\right)+\lambda_1\left(y-c\right)=0\ and\left(y-c\right)+\lambda_2\left(y+c\right)=0$

$\Rightarrow-\lambda_1=\frac{y+c}{y-c};\frac{-1}{\lambda_2}=\frac{y+c}{y-c};\lambda_1\lambda_2=1$

Locus is $\left(\frac{y-ax}{z-c}\right)\left(\frac{y+ax}{z+c}\right)=1$

$a^2x^2-y^2+z^2-c^2=0$

88. Equation of the plane π is $A\left(x-1\right)+B\left(y-1\right)+C\left(z-1\right)=0$ where A + 0, B–1, C=0 and –A+B+0.C=0

$\Rightarrow A=B=C$ and the equation of π is X + Y + Z = 3x. which meets the coordinate axes at A (3, 0, 0), B (0, 3, 0) and C (0, 0, 3). So volume of the tetrahedron

$$OABC = \begin{vmatrix} 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 1 \\ 0 & 3 & 0 & 1 \\ 0 & 0 & 3 & 1 \end{vmatrix} = \frac{1}{6} \times 27 = \frac{9}{2}$$

(B) $AB = BC = CA = 3\sqrt{2}$, so the area of the face ABC which is an equilateral triangle

$= \frac{\sqrt{3}}{4} \times \left(3\sqrt{2}\right)^2 = \frac{9\sqrt{3}}{2}$. Each of the face OAB, OBC, OCA is an isosceles triangle with sides 3, $3\sqrt{2}$

So area of these faces is $\frac{1}{2} \times 3\sqrt{2} \times \sqrt{9 - \frac{9}{2}} = \frac{9}{2}$

(C) required distance is $\left|\frac{3\sqrt{3} + 3\sqrt{3} - 3 - 3}{\sqrt{1+1+1}}\right| = \frac{9}{2}$

(D) Distance of O from the face $ABC = \left|\frac{0+0+0-3}{\sqrt{1+1+1}}\right| = \sqrt{3}$ distance of A, B, C from opposite faces

OBC, OCA, OAB are equal and is 3.

89. Using the concept of family of planes let required equation of plane is

$p_1 + \lambda p_2 = 0,$ since the plane passes through (3, 2, 1) therefore

$\Rightarrow 2x - y + z - 2 + \lambda\left(x + 2y - z - 3\right)$

$\Rightarrow \lambda = -1.$

Required plane is $x - 3y + 2z + 1 = 0$

90. Equation of the plane passes through $(-1, 3, 2)$

is $a(x+1) + b(y-3) + c(z-2) = 0 \ldots(1)$

(1) is perpendicular to $L = 0,$

(Dr's of L are $(2i - j + k) \times \left(i + 2j - \hat{k}\right)$) we get

$\frac{a}{-1} = \frac{b}{3} = \frac{c}{5}$ therefore equation of plane is

$\left(\vec{r} - \left(-\hat{i} + 3\hat{j} + 2\hat{k}\right)\right).\left(\hat{i} - 3\hat{j} - 5\hat{k}\right) = 0$

91.

92. Let $P(\alpha+3,-2\alpha+5,\alpha+7)$ and

$Q(7\beta-1,-6\beta-1,\beta-1)$ be the points on the

given lines so that PQ is the line of shortest distance

between the given lines D.r's of $PQ=(\alpha-7\beta+4,-2\alpha+6\beta+6,\alpha-\beta+8)$

Since PQ is perpendicular to the given lines

$1(\alpha-7\beta+4)-2(-2\alpha+6\beta+6)+1(\alpha-\beta+8)=0$

$\Rightarrow 6\alpha-20\beta=0 \Rightarrow 3\alpha-10\beta=0$ ....(1)

and,

$\Rightarrow 7(\alpha-7\beta+4)-6(-2\alpha+6\beta+6)+1(\alpha-\beta+8)=0$

$\Rightarrow 7\alpha-49\beta+28+12\alpha-36\beta-36+\alpha-\beta+8=0$

$\Rightarrow 20\alpha-86\beta=0 \Rightarrow 10\alpha=43\beta$.....(2)

From (1) and (2) $\alpha=0,\beta=0$

$P=(3,5,7),Q=(-1,-1,-1)$

Distance between PQ is $=2\sqrt{29}$

D.r's of $PQ=(-4,-6,-8)=(2,3,4)$

Equation of PQ is $\frac{x-3}{2}=\frac{y-5}{3}=\frac{z-7}{4}$

(I) Let the image of (2, 1, 4) in the plane

$2x-y+z+5=0$ be (x_1,y_1,z_1) then

$$\frac{x_1-2}{2}=\frac{y_1-1}{-1}=\frac{z_1-4}{1}=\frac{-2\{2\times 2-1+4+5\}}{(2)^2+(-1)^2+(1)^2}=-4$$

$x_1=-8+2=-6; y_1=4+1=5, z=-4+4=0$

$\therefore$ The image is (-6, 5, 0) = $(\alpha_1,\alpha_2,\alpha_3)$

II) Since the image point of (2, 1, 4) is $(-6,5,0)$, the mid point of them is the foot of the perpendicular

i.e., (-2, 3, 2) = $(\beta_1,\beta_2,\beta_3)$

93. (I) Let the image of (2, 1, 4) in the plane

$2x-y+z+5=0$ be (x_1,y_1,z_1) then

$$\frac{x_1-2}{2}=\frac{y_1-1}{-1}=\frac{z_1-4}{1}=\frac{-2\{2\times 2-1+4+5\}}{(2)^2+(-1)^2+(1)^2}=-4$$

$x_1=-8+2=-6; y_1=4+1=5, z=-4+4=0$

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(II) Since the image point of (2, 1, 4) is $(-6,5,0)$, the mid point of them is the foot of the perpendicular

i.e., (-2, 3, 2) = $(\beta_1,\beta_2,\beta_3)$

94-95. $L_2: x-3y-4=0 \;\&\; 4y-z+5=0$

$P\equiv(4,0,5)$ $\qquad \overrightarrow{DR}=\vec{n}=\overrightarrow{n_1}\times\overrightarrow{n_2}$

$\vec{n}=3\hat{i}+\hat{j}+4\hat{k}$ $\qquad \begin{pmatrix}\vec{n}_1=\hat{i}-3\hat{j}\\ \vec{n}_2=4\hat{j}-\hat{k}\end{pmatrix}$

$L_2:\frac{x-4}{3}=\frac{y}{1}=\frac{z-5}{4}$ ….(1)

$P_1: x+y-z=5,\; L_2:(3\lambda+4,\lambda,4\lambda+5)$

$$\frac{x-(3\lambda+4)}{1}=\frac{y-\lambda}{1}=\frac{z-(4\lambda+5)}{-1}=-2\left(\frac{-6}{3}\right)=4$$

$(x,y,z)\equiv(3\lambda+8,\lambda+4,4\lambda+1)$

Image of line $L_2:\frac{x-8}{3}=\frac{y-4}{1}=\frac{z-1}{4}$

$P_1=0; P_3+0; P_1+\lambda P_3=0$

$(x+y-z-5)+\lambda(x-3y-4)=0$

$x(1+\lambda)+y(1-3\lambda)-z-(5+4\lambda)=0$

$p_1=p_2$ from (4, 0, 0)

$$\Rightarrow\left|\frac{1}{\sqrt{(\lambda+1)^2+(1-3\lambda)^2+1}}\right|=\left|\frac{1}{\sqrt{3}}\right|$$

$\lambda=0,\frac{2}{5}$

$P: 7x-y-5z-33=0$

$L_1: x+y-z=5, 2x-y+\lambda z=3$

Point $\equiv\left(\frac{8}{3},\frac{7}{3},0\right)$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -1 & \lambda \end{vmatrix} = (\lambda - 1)\hat{i} - (\lambda + 2)\hat{j} - 3\hat{k}$$

If line L_1 and L_2 are coplanar

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{4}{3} & \frac{-7}{3} & 5 \\ 3 & 1 & 4 \\ (\lambda-1) & -(\lambda+2) & -3 \end{vmatrix} = 0$$

$$\lambda = -\frac{5}{4}$$

96-98.

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0 \Rightarrow \alpha = 4$$

Put $z = 0$ in P_1 and P_2

$2x + y = 1$ and $x - y = 2$

$\therefore \quad P = (1, -1, 0)$

Put $x = 0\, in P_1 and P_2$

$y + z = 1$ and $-y + z = 2$

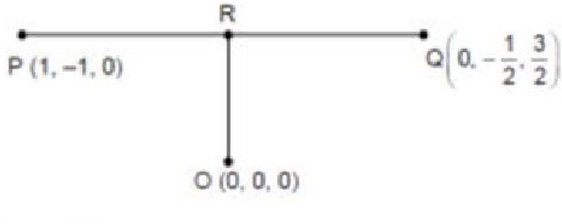
$\therefore \; Q = \left[0, -\frac{1}{2}, \frac{3}{2}\right]$

$\vec{PQ} = -\hat{i} + \frac{1}{2}\hat{j} + \frac{3}{2}\hat{k}$

Projection of PQ on x – axis = $\left|\vec{PQ}.\hat{i}\right| = 1$

Equation of line PQ is

$$\frac{x-1}{-1} = \frac{y+1}{1/2} = \frac{z}{3/2} = \lambda$$



Let R = $\left(-\lambda+1, \frac{\lambda}{2}-1, \frac{3\lambda}{2}\right)$

$\vec{OR}\vec{PQ}=0$

$\Rightarrow -1(-\lambda+1)+\frac{1}{2}\left(\frac{\lambda}{2}-1\right)+\frac{9\lambda}{4}=0$

$\Rightarrow \lambda=\frac{3}{7}$

$\therefore \qquad R=\left(\frac{4}{7},-\frac{11}{14},\frac{9}{14}\right)$

$\Rightarrow 7a+14b+14c=2$

Area = $\frac{1}{2}\left[\vec{PQ}\times\vec{OQ}\right]=\sqrt{\frac{19}{16}}$

$\therefore \qquad$ a – b = 3

99-100.

$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$ taking dot product with $\vec{a}$ + 0 +[a b c] = $p\vec{a}.\vec{a}+q\vec{a}\vec{b}+r\vec{a}.\vec{c}$

$[abc]=p+\frac{q}{2}+\frac{r}{2}$ ……………..(1)

Similarly taking dot product with $\vec{b}$

$0=\frac{p}{2}+q+\frac{r}{2}$ ……………..(2)

And taking dot product with $\vec{c}$

$\left[\vec{a}\vec{b}\vec{c}\right]=\frac{p}{2}+\frac{q}{2}+r$ …………..(3)

From equation (1) and (3), we get p = r

From equation (2) p = -q

$\Rightarrow$ [a b c] = $p-\frac{p}{2}+\frac{p}{2}=p$

But we know that $\left[\vec{a}\vec{b}\vec{c}\right]^2=\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c}\end{vmatrix}=\begin{vmatrix}1 & \frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & 1 & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2} & 1\end{vmatrix}$

$\Rightarrow \left[\vec{a}\vec{b}\vec{c}\right]^2 = \frac{1}{2}$

$\Rightarrow \left[\vec{a}\vec{b}\vec{c}\right]^4 = \frac{1}{4} = 0.25$

$p = \frac{1}{\sqrt{2}}, q = -\frac{1}{\sqrt{2}}; r = \frac{1}{\sqrt{2}}$

Now, $20p^2 + 21q^2 - 22r^2 = \frac{20}{2} + \frac{21}{2} = 9.50$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